MILITARY SPECIFICATION

SIGNAL, UNDERWATER SOUND (SUS)

MARK 84 AND MODS

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 General - The devices covered by this specification shall provide one-way communication from an aircraft in flight to a submerged submarine or simulate an ASW weapon launch. The secondary functions are to provide an active acoustic reference, and to assist in determining misdistance from submarine in simulated weapon drops. The devices have four or more manually selectable acoustic codes in the audible frequency range. The devices are electromechanical, continuous wave, frequency shift coded, expendables. The device will be referred to as the "signal" in this specification.

1.2 Classification - The signal shall be of the following types:

Type	Description (See Figure 12)
MK 84 MOD 0	Four codes (1 through 4)
MK 84 MOD 1	Five codes (1 through 5) Operationally rain and spray tight.
MK 84 MOD 2	Five codes (1 through 5) Operationally rain and spray tight. 14" suspension provision.

* 1.3 Associated Equipment - The signal shall be capable of detection and code identification with submarine sonars (in the signal's frequency ranges), underwater telephone AN/UQC, and sonobuoys in the AN/SSQ-41 and 57 series. The signal shall be capable of being launched from bomb rack or shackle adapters such as the ADU-369/A; from bomb racks or shackles with 14" suspension (MOD 2 signal only); from sonobuoy launcher-containers (SLC) capable of using a split liner (sabot); and manually through aircraft free-fall chutes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5845

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

	MIL-A-8591	Airborne Stores and Associated Suspension Equipment; General Design Criteria for
*	MIL-P-23312	Pallet, Material Handling, Metal (for Ordnance Items), MK3 MOD 0, MK12 MOD0 and MOD 1.
	MIL-R-81597	Rack, Bomb, MAU-38/A
*	WR-54/131	Weapons Requirement, Palletizing Fleet Issue Unit Load; Signals, Underwater Sound MK57, MK61, and MK64, or MK59 MOD 0 in Small Arms Ammnition Box MK1.
*	WR-54/167	Weapons Requirement, Palletizing Fleet Issue Unit Load; Signals, Underwater Sound MK9, MK57, MK61, MK64 MOD 0 and MK59 MOD 5 in 20MM Ammunition Box MK3 MOD 3.

STANDARDS

Federal

Federal Test Method STD No. 101	Preservation, Packaging, and Packaging Materials; Test Procedures
FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-648	Design Criteria for Specialized Shipping Containers
MIL-STD-810	Environmental Test Methods
* MIL-STD-1168	Ammunition Lot Numbering
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities

DRAWINGS

* Bureau of Naval Weapons

2302052	Tail Fin

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* 2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 1051, 5801 Tabor Avenue, Philadelphia, PA 19120. The drawings may be obtained from the Cognizant Field Activity for the signal; see address in contract.

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI S1.20 Calibration of Underwater Electroacoustic Transducers

(Application for copies should be addressed to: American National Standards Institute, Attn: Sales Dept., 1430 Broadway, New York, NY 10018.

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General - The signal shall be as specified herein. The signal is a low cost, mass produced, expendable item, designed for one-time use, and is not recovered or refurbished after use. There is no upkeep during the signal's life. The signal is usually expended at the end of an aircraft ASW training exercise where high cost sensors were expended and failure of the signal to function negates scoring, readiness, and the cost that was invested in aircraft, submarine, and their crews operating time. It may also be used in communications, the failure of which could impact mission safety for a submarine. For these reasons, sound construction and reliable functioning are mandatory. Tests are referenced at the end of paragraphs by Table and Sequence or Item Number. Tables and Figures are located following Section 6.

* 3.2 Classification of Requirements - The requirements for the signal are classified herein as follows:

Requirements	Paragraph
First Article	3.3
Change in Design	3.4
Parts and Material	3.5
Design and Construction	3.6
Operational Environment	3.7
Performance	3.8

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* 3.3 First Article - When ordered in the contract, a preproduction sample of signals shall be manufactured, using the methods and procedures intended for use on production signals, and shall successfully be tested as specified under the quality assurance provisions herein prior to a release to production (4.4).

* 3.4 Change in Design - All changes in design, that may affect the performance of the signal, shall be qualified prior to use in production, by test of signals manufactured with the changes incorporated, as specified under the quality assurance provisions herein (4.11).

3.5 Parts and Material - The first consideration in the selection of parts and materials shall be the reliable fulfillment of the major performance objectives. The second consideration, for use in an expendable, shall be low cost. Degradation of materials is acceptable, if the part's function is not impaired, within the specified life of the signal. The following general requirements apply:

(a) Controls: Shall be of rugged design and construction that will not be damaged when repeatedly operated by unskilled personnel under service conditions. (See I-1.6.J, II-1)

(b) Dissimilar metals: Shall have their interface surfaces protected to prevent electrolytic corrosion and exposed surfaces shall be stress-corrosion resistant for the specified signal life. (See I-1.6.La)

(c) Castings: That form a watertight barrier shall be free of porosity or protected by impregnation only to the extent of insuring freedom from water within the barrier during functioning until after electrical termination. (See I-3.Aa, Ab, Ac)

* (d) Fungus resistance: All exposed parts shall be fungus resistant in all locations that affect the function of the signal for a period not less than 60 days for the MOD 0 and one year for the MODs 1 & 2 after removal (exposure) from the long term Shelf Life stowage container. See 3.7.4(b) for additional requirements. (See I-1.6.La)

(e) Explosive fumes: Shall not be liberated when in the non-functioning operational environment.

* (f) Reliability: The parts used must be equal to or more reliable than those qualified under First Article or Change in Design. Parts and material of high failure rate and of limited shelf life shall not be used. This shall be effected by an evaluation of each element of the design.

* (g) Flammable materials: The assembled signal shall be self-extinguishing and not subject to explosion. Fire retardant additives, if used, shall be a permanent part of the materials. The transducer boot and surface marking materials are excepted from the flammability requirements. (See I-1.6.Lb)

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* (h) Design: Shall be accomplished by the contractor, subject to the requirements of this specification. The design layout and the assembly of the piece parts, components, and assemblies shall be such as to facilitate quantity production; and should be capable of reworking or replacement in the event that resubmission is required. Potting or other materials used as fillers shall not be subject to shrinkage voids or size variations that would cause an erratic trajectory or contribute to component or circuit damage at high spin rates and high water impact shocks.

(i) Microphonics: Shall not affect the operation.

* (j) Exposed plastics: Shall be of grades that are resistant to change in properties such as result in embrittlement, stress cracking, swelling, or softening when exposed to cleaning solvents, marking removal solvents, hydrualic liquid, oils, grease, and fuel for short or limited periods; and for prolonged exposure to marking inks or paint, and to aging. As an alternative, when using a plastic susceptible to the foregoing defects, except aging, a durable preventative coating may be employed. Alternative marking may be made on other surfaces to be visible through the plastic (See Figure 2); applied over a barrier coating; or a compatible marking, ink or paint, applied directly, providing that the design data package and fabrication process sheets provide for a compatible cleaning solvent for fabrication and/or change in marking. Surface treatment or coating, if required, shall insure that plastic defects will not occur during a one year Service Life exposure (3.8.1), as a result of direct handling by personnel with grease and/or oil on their hands. (See I-1.6.La)

(k) Explosive Devices or Actuators: Shall not be used. (See I-1.6.La)

* (l) Water Impact Resistance: Material shall be capable of withstanding water impact forces resulting in dynamic crushing, punching, shearing, torque, pressure, and adiabatic heating of trapped air resulting from drops from the speed/altitude envelope indicated in Figure 6. To withstand this, the material shall be of adequate cross-section and free from unsupporting voids or the voids shall be rigidly controlled. (Reference 3.7.9 and 6.7.3)

* (m) Recycled, virgin, and reclaimed materials: May be used, but shall be restricted to the same proportions and processing used in the First Article sample that qualified, and shall be so identified with specific data on the configuration control drawings (See 6.3).

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* 3.5.1 Standard and Nonstandard Parts and Material Qualification - All parts and material shall be qualified by performance testing of the signal after exposure to all storage and during any of the service and operational environments during First Article. Parts and material from alternate sources shall be likewise qualified, as specified under the quality conformance section, by the contractor, except those parts classified as Critical (4.5). Critical parts and material from alternate sources shall be tested during original qualification of the signal at First Article or during requalification of the signal for Design Change Approval prior to their use in production (4.11). Nonstandard parts approval, other than the above, is not required.

3.5.2 Critical Parts and Material - The following parts and material are designated Critical:

(a) Electronic parts specially made by another contractor/vendor for the signal (non-stock) such as; sea water batteries and transducers; and

(b) All parts or material the contractor does not have specification control of or does not procure to the contractor's specification control drawing.

3.6 Design and Construction - The signal shall be designed and constructed to perform as specified herein in any of the extremes of the operational environment after enduring the extremes of various storage environments for the periods designated. The design shall incorporate all the features detailed in the following and those unique to the MOD ordered (1.2)(6.2(b)).

3.6.1 Form Factor - Shall be (See I-1.6A):

(a) Diameter: 76.2 $\pm$ .762mm (3 $\pm$ .030 inches) at the nose, body, and tail shroud bearing surfaces. 1° of draft of the bearing surfaces shall not be exceeded. (See II-7)

* (b) Diameter, Nonbearing Surfaces: The variation of the body diameters at the transducer cylinder and tail pipe or cgive, taper, or step shall not exceed the bearing surfaces; otherwise they are optional within the specified performance limitations, except bumps or lumps under the transducer boot from wiring connection shall be of low (not sharp) profile. No portion of the transducer boot OD shall extend beyond the actual nose and mid-body bearing surfaces line. (See II-7)

(c) Length: The overall length of the signal shall be 381 + .000 -2.286mm (15.000 + .000 - .090 inches) from the flat end of the nose to the outside edge of the tail shroud. (See II-7)

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* (d) Length, Longitudinal Surfaces: The nose shall have a 12.7 + 1.524mm (.500 ± .060 inches) radius. The nose bearing surface shall extend a minimum of 47.625mm (1.875 inches) from the nose face. The tail shroud bearing surface shall extend a minimum of 44.45mm (1.75 inches) from the tail face. The mid-body bearing surface shall extend a minimum of 38.1mm (1.5 inches). The transducer shall be located between the nose and mid-body bearing surfaces. (See II-7)

* (e) Nose: The end face shall not have switches, legend, or other components extending or raised above the end face; and all insets shall be symmetrical or compensated to insure even air/water flow characteristics; otherwise it shall be flat.

* (f) Tail: The end face shall be flat. The shroud shall enclose four canted fins. (Reference: BUWEPS Dwg 2302052)

(g) Reference: Figure 1 illustrates (a) through (f) above.

* 3.6.2 Weight - The signal weight shall cause a sink rate of 4.42 ± 0.15m/sec (14.5 ± 0.5 ft/sec) but shall not exceed 3.175 kg (7 lbs). See 3.8.3. (See I-1.6B & Lc)

* 3.6.3 Center of Gravity - The center of gravity shall be located to cause the signal to stabilize nose first with the longitudinal axis conformed to the in-air trajectory, when launched from the altitudes and speeds in Figure 6, with the signal fully stabilized to the trajectory at water entry, and shall not exceed the maximum dimension in Figure 1. See 3.8.3. (See I-1.6B)

3.6.4 Finish - The exterior surfaces shall be treated as necessary to retain all required marking for the period specified for shelf life. (See I-1.6I)

* 3.6.5 Marking - The signal shall be marked as follows, on the body, or to be seen through the body sides, in black 4.76mm (3/16 inch) Arabic numerals and Gothic or Futura or similiar type capital letters, parallel to the longitudinal axis, except as specified otherwise: (Reference: Figure 2). (See I-1.6C, III-9)

(a) SIGNAL, UNDERWATER SOUND, MK 84 MOD___

* (b) See (c).

* (c) National Stock Number (NSN) followed by a space and the Naval Ammunition Logistic Code (NALC) (e.g., 1360-00-052-1480 SW06 (for the MOD 0) or 1360-01-037-0588 SW39 (for the MOD 1)).

* (d) FSCM_____MFR P/N_____(e.g., FSCM 88888 MFR P/N DL 33421-9) the configuration control number; in this case the number of the Data List that lists all the drawings and specifications of the design and their latest revisions, the final group of numbers or letters after the dash representing the top design revision for the signal.

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(e) WT_____KG/_____LBS (nominal to one place; e.g., 3.2 KG/7 LBS)

* (f) _____(manufacturer's name).

(g) _____(contract number).

* (h) LOT_____(e.g., LOT MRI79D001-001). In accordance with MIL-STD-1168. (NOTE: This may be stamped in by hand, and may be irregular, but must be legible.)

(i) SUS MK____MOD____(shall be a minimum of 12.7mm (.5 inches) in height, located circumferally around center of the tail shroud to read when the signal is standing on its nose.)

* (j) The code setting switch control shall be marked, shaped, or have cast in an arrowhead or other index of unambiguous meaning that will indicate the code setting selected.

* (k) The codes 1, 2, 3, etc. shall be marked or cast on the surface facing the detented code switch setting positions. There shall be no marking indications or components of assembly that could give an incorrect meaning/position. Alignment shall be centered ± 1/4 of the width, on the number and or its index mark. The size and contrast of the code numbers, positions, and indexes shall be clearly readable under reduced light conditions, and by flashlight equipped with a red lens at night. Reference MIL-STD-1472. (NOTE: The location of (j) and (k) above may be exchanged.) (See I-1.6J)

* (l) NONEXPLOSIVE (in 9.5mm (0.374 inch) minimum letters) in one line.

* (m) US (in 9.5mm (0.374 inch) minimum letters).

(n) TRANSDUCER (shall be located in the middle of the transducer to be read when the signal is stood on its nose; it shall be repeated to be visible on all sides; the marking shall contrast from the background and shall not defeat, harm, or degrade the transducer covering; marking irregularities are acceptable, but marking must be legible.)

* (o) A minimum of the 44.45mm (1.75 inch) external tail shroud bearing surface shall be blue in color equivalent to color #25109 of FED-STD-595.

(p) Any part that must be manually removed from the signal, prior to its use, shall have its assembled exposed portion bright red in color and the instructions in 3.175mm (.125 inch) minimum height letters for its removal marked or cast in the part or a decal fixed to the part, or such marking adjacent or tagged thereto if the area available is too small. (See I-1.6Ld)

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* 3.6.6 Test Provisions - The assembled signal shall have provisions for applying external power and testing all codes by code switch control (3.6.9). The assembled signal shall also have provisions for testing for leaks in all watertight, vaportight (hermetic seal), and sealed but not vaportight compartments appropriate to the direction and pressures that are applied under all environments including Shelf and Service Life, captive and free flight, and underwater trajectories. The leak test capability shall take into account low pressure leakage before seals are set or expanded, and subsequent higher pressures. The provision is not limited and may include pressure, vacuum, immersion, gas and gas tracer and other techniques providing that the provision will not be a potential leakage point or result in deleterious material attack. The provision for external power application may be combined with the provision for leak test(s). Prior to final sealing, all internal volumes shall be flushed with inert gas or other means taken to ensure a dry internal atmosphere that will not cause material or part deterioration. (See I-1.6.D, I-6, II-6)

* 3.6.7 Provisions for Reworking - See 3.5(h) and 6.7.1.

* 3.6.8 Watertight and Vaportight Integrity - The MOD 1 & 2 signals shall remain watertight from rain, melt (from ice, frost, snow) and salt spray; and shall retain a water vapor transmission rate (WVTR) that does not exceed the desiccant capacity for the Service Life period under the conditions specified (3.6.12(b)) or shall be hermetically sealed until water entry (including captive and free flight). The parts of the signal required by design to be watertight during the period of acoustic output, after water entry, shall continue to be watertight until output termination and until reactivation is precluded by circuit or power source destruction, or other positive means that would insure that leakage or other malfunction would not reactivate an output of any form (all MODs). (See I-1.6.Eb; I-3.Aa, Ab, Ac, B; I-7; II-3 & 6)

* 3.6.9 Code Switch - The code select switch control location is optional providing that it is visible and can be set by hand or by a hand-held coin the size of a U.S. quarter. The control shall be rotorary with Code ONE position in the extreme counterclockwise position. The codes shall be selectable sequentially in increasing code numbers clockwise. Each code position shall be detented to insure that an intermediate position will virtually be impossible to set and water entry forces will not index the switch to another code. The switch control shall not be susceptable to accidently changing codes by handling, such as would be the case if the tail shroud was the manual control element and did not have an added position latch. The code positions shall be easily felt by the setter. The maximum setting torque shall not exceed 0.565 Nm (5 inch-pounds) and maximum breakaway torque from the stored position shall not exceed 1.13Nm (10 inch pounds). The switch shall have a positive stop adjacent to the highest and lowest code settings. The switch circuit shall be so designed that accidental switch positioning between code positions or switch contact failure will not result in signal output on any code. (See I-1.6.A & J, II-5, IV-4)

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3.6.10 Suspension Lugs (MOD 2 only) - The signal shall be capable of suspension on a bomb rack or shackle having 355.6mm (14 inches) between hook centers. Suspension lugs shall be self-housing within the body and tail structure to not affect air and underwater trajectories. The housing force shall be adequate to hold or grip the signal up to the bottom of the bomb rack to prevent swaying during flight and flutter from air turbulence when the bomb bay is open. Design to 14 g shock load without deformation and 21 g shock load without failure, deformation acceptable between 14 and 21 g's. Lugs shall have sufficient constructed cross section to ensure not falling through the clearance remaining after bomb rack or shackle hooks are latched. Provision shall be made to assist in attaching lugs to bomb rack hooks. This may be accomplished with simple tools, pins, or a releasable assembly position in the lug mechanism. The signal shipping configuration shall not exceed the dimensions specified under form factor. Reference: MIL-A-8591, MIL-R-81597, and Figure 3. (See I-1.6.Ld, II-4, IV-5)

3.6.11 Keepers - Devices may be incorporated in the design that will retain spring loaded suspension lugs or watertight integrity components while handling and/or loading. The requirement for removal of the devices shall be obvious by the design, such as ring and tag or other indicator. Shipping configuration shall not exceed the dimensions under form factor. (MODs 1 or 2 as applicable) (See I-1.6.Ld)

* 3.6.12 Power Source - The power source shall be self-contained and activated by salt water at water entry, or by water entry (3.7.9) and shall not require charging or replacement. Power sources requiring sea water flooding shall be configured as follows: (See I-6)

(a) hermetically sealed in a water and vapor tight compartment. See 6.5 (MODs 1 & 2); or

(b) Sealed in a watertight but not necessarily water/vapor-tight compartment. The compartment shall contain a sufficient quantity of desiccant to initially dry the internal air volume and maintain dry air for a minimum of one year direct exposure to 26.7°C (80°F), at 80% R.H., the Service Life period and condition. The water vapor transmission rate (WVTR) of the permeable compartment boundaries and seals, the area of transmission, the daily weight of desiccant water absorption, and the total desiccant absorption capacity by weight shall be detailed in the configuration control drawings. The compartment shall have a non-reversable humidity indicator visible through a window. The indicator shall indicate within a 24 hour period when exposed to 80 $\pm$ 2.5% R.H. at a reference temperature of 23.9°C (75°F). The indicator size shall allow for reading (color run) and the window shall allow for seeing the color run under white light. The design of the signal shall facilitate replacement of the power source and desiccant, at depot level, following exposure with indication. This provision shall not be interpreted to require any sort of plug-in receptacle, or other facility, but shall be designed to enable the replacement without destruction of other parts. Lead splicing is acceptable for replacement purposes. (MODs 1 & 2); and (See I-7&8, II-3, III-1, IV-3)

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(c) Shielded from physical damage (all MODs).

(d) In the designs that require the shock from signal impact to initiate functioning, the signal shall also be capable of being manually prepared for operation prior to hand launching at the surface. Instructions shall be included in the instruction sheet provided (6.3).

3.6.13 Human Engineering - The devices that personnel must set, assemble, or handle shall be simple in design and not difficult or unhandy for a person with no disability and without gloves to accomplish. The markings shall be legible, and instructions shall be within the comprehension of the ninth grade level of U. S. public education. (See I-1.6.K)

3.6.14 Reliability - The signal shall function as specified with a 96% or better reliability.

* 3.6.15 Safety - The signal shall not be capable of causing injury to personnel during unpackaging, assembling, setting, loading; or during down-loading and return to storage condition; or during manual launching. The signal and/or its appendages shall have no sharp edges or ends that can cut hands. There shall be no restrained power source incorporated in the design other than a battery(s), retracting springs for snap down lugs, and springs used to detent and secure the setting switch, except as may be incorporated in accordance with the approval of the NAVAIRSYSCOMHQ technical code, and contingent on confirming safety tests. No part of the signal shall separate or deploy, on release from the aircraft, by design or by crew error that would be capable of fouling or damaging the aircraft. Restrained power sources prohibited shall include electrostatic charges capable of causing involuntary reactive body movement of personnel handling the signal. (See I-1.6.K)

3.6.16 Workmanship - The signal shall be fabricated and assembled without defects or damage induced that could make the part (or signal) unsatisfactory for its intended use. The procedures used shall be consistent with obtaining identical parts or assemblies from unit to unit. This shall specifically be applicable to selection of good or whole parts, securing and dressing each part in assembly for extremely high shock, assembly of all watertight devices, and quality conformance testing. Neatness is not of primary concern, but all workmanship that would lead to hidden or latent defects are; in this regard there shall be a minimum of dust, dirt, or lint on or in the vicinity of parts or assemblies. Care shall be exercised as to the shielding afforded adjacent dust, grit, liquid, or caustic fume producing operations. All parts shall have been cleaned to remove residue incident to fabrication, such as: oils, release agents, sand, fluxes, wire edges, etc. All threaded devices shall be started in a manner that will insure that cross-threading cannot happen. All wires exposed to water under pressure, shall be carefully handled to insure that insulation nicks or pin holes are not induced; this shall include use of wire feeding mechanisms and hand tools that are free of defect causing systems or conditions. All sealing mechanisms such as o-rings and their bearing surfaces shall be lubricated as specified in the design and assembled carefully to avoid cutting or pinching, twisting or other damage. (See I-1.6.K, II-2)

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3.7 Operational Environment - The signal shall perform satisfactorily after enduring the non-operating environments of transportation, storage, handling, captive flight, free fall, and water entry; and shall perform satisfactorily in the underwater operating environment. The signal shall survive any combination of environmental extremes that could logically happen together in nature and in service prior to and during the ASW training cycle.

3.7.1 Primary Transportation - The signal, protected by a complete packaging system, blocked and braced, in all modes of transportation by common carrier, shall survive shipment a minimum distance of 4800 kilometers (3000 miles) on land and 9300 kilometers (5000 nautical miles) on the sea or in the air. (Reference: MIL-STD-810) (See I-1.I)

* 3.7.2 Terminal Transportation - The signal, protected by its unit container (when required) and its exterior container, unrestrained, shall survive shipment a minimum distance of 81 kilometers (50 miles) on land by truck. (Reference: MIL-STD-810) (See I-1.2.F)

3.7.3 Long Term Storage - The signal shall survive a shelf life of ten years when protected by a complete packaging system, stored under cover, and experiencing moderate daily temperature cycling.

3.7.4 Short Term Operational Storage - The signal shall survive, in sequence:

* (a) The balance of long term storage (10 year Shelf Life) when maintained hermetically sealed or in the hermetically sealed unit container, under cover:

(i) With extreme temperature cycling on land, or; (Reference: MIL-STD-210, MIL-STD-810) (See I-1.6.Da)

(ii) With moderate temperature on a ship with vibrations up to 50 Hz. (Reference: MIL-STD-210, MIL-STD-167) (See I-1.I)

(b) When unprotected, oriented at any attitude, and under cover, a minimum shelf life of: 60 days for the MOD 0 and one year for MODs 1 and 2:

(i) In a warm, salt air atmosphere as encountered on a surface ship, or: (Reference: MIL-STD-210) (See I-1.6.Ea)

(ii) In a hot, dry, and dusty atmosphere experienced in a desert, or; (Reference: MIL-STD-210) (See I-1.6.Da & Ec)

(iii) In a warm, humid, and fungus producing atmosphere experienced in the moist tropics not exceeding the conditions in Figure 4, or; (Reference: MIL-STD-210) (See I-1.6.Db)

(iv) In a very cold, dry atmosphere that would be experienced in the sub-artic. (Reference: MIL-STD-210) (See I-1.6.Da & Dc)

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3.7.5 Handling - The signal, unprotected, shall survive, with minor damage, a free drop of 1.22 meters (4 feet) on its side or ends, at any angle, on a flat hard surface such as a steel deck or concrete floor. Damage sustained shall not cause premature operation or affect normal operation. (See I-1.6.Fa, II-3).

3.7.6 Captive Flight - The signal, unprotected, shall survive in-flight conditions in ASW aircraft which include vibration levels up to 5g's, and arrested landing shock loads up to 14g's, and the following: (See I-1.6.Fb & Fc)

(a) Unsecured at cabin conditions, or; (Reference: MIL-STD-810)

* (b) Suspended from a bomb rack or secured in a bomb rack adapter or secured in a sonobuoy launcher-container in split liners or sabot and exposed to external temperature/altitude conditions from -65°C (-85°F) at 12,200 meters (40,000 feet) to +40°C (+104°F) at the surface and the intervening temperature/altitude combinations within the envelope in Figure 5 and up to +40°C (+104°F) residual temperatures in initial accent up to 12,200 meters (40,000 feet) altitude. Additionally, exposure on both wing and open bomb bay rack stations including drenching rain without premature functioning of the signals (MODs 1&2), and to extreme air turbulance. (Reference: MIL-STD-210 and MIL-STD-810). (See I-1.6.Da, Eb, Fb; I-2.Aa, Ab, Ac)

* 3.7.7 Launch - The signal shall withstand launching forces in the longitudinal axis up to a maximum of 150g's in either direction without premature functioning when launched from an aircraft within the speed/altitude envelope of Figure 6. The signal shall activate at water entry when launched from the speed/altitude envelope of Figure 6. See 3.6.12(d). (See I-5)

* 3.7.8 Free Fall - The unprotected signal (MODs 1 & 2) shall survive the fall through drenching rain and reduced temperatures without premature functioning from the launch altitude to water entry, Figures 5 & 6, a period of less than 1 1/2 minutes. (See I-1.6.Eb)

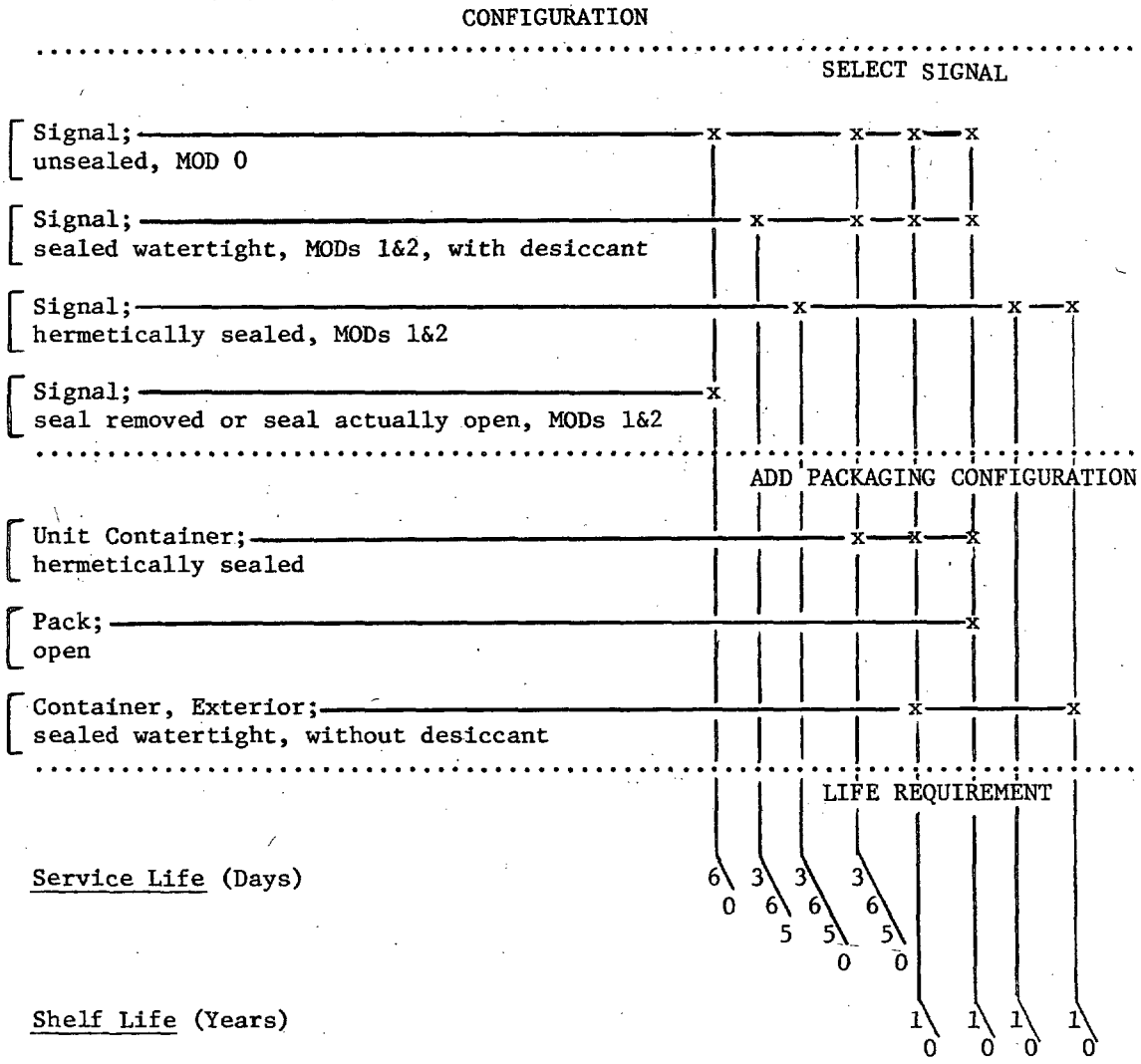
3.7.9 Water Entry - The unprotected signal shall survive the water entry shock. (Reference: 6.7.3)

* 3.7.10 Underwater - The signal shall function in accordance with 3.8.7 in sea water 1.5 to 3.6 percent salt by weight, within the temperature range of 0° to +35°C (+32° to +95°F), and pressure increase occurring from descent until specified signal acoustic output termination. (See I-2.Aa, Ab, Ac)

3.8 Performance

* 3.8.1 Life - The signal shall have the long term Shelf Life and short term Service Life indicated below when subjected to the environments specified in 3.7.3 and 3.7.4. All parts and materials incorporated shall not change their performance characteristics during these periods. Where necessary, parts or components that deteriorate, malfunction, or expend themselves in the presence of moisture shall either be hermetically sealed, or sealed such that the material WVTR is offset by desiccant moisture pickup and/or the WVTR is otherwise reduced to obtain the performance. See 3.6.8.

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3.8.2 Suspension - The suspension lugs shall hold the signal compressed against the bottom of the bomb rack, and restrain all side and fore and aft movement, but shall not prevent bomb rack operation. The lugs shall house themselves immediately after bomb rack release of the signal (MOD 2 only) (See I-1.6.Ld)

* 3.8.3 Trajectory - The signal in-air and in-water trajectory shall be consistant from signal to signal. The in-air trajectory and in-water trajectory shall be the same in relationship as the densities of the two mediums. The in-water trajectory shall not vary more than 152.4mm (6 inches) in 30.48mm (100 feet) in a still fresh water reference. See 3.6.2 for sink rate and 3.6.3 for stabilization. (See I-1.6.Lc; I-3.Aa, Ab, Ac)

* 3.8.4 See 3.8.3

* 3.8.5 See 3.8.3

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* 3.8.6 Activation - The signal, when pre-stabilized at any temperature between -20°C (-4°F) and +40°C (+104°F) before water entry, shall propagate sound in accordance with 3.8.7 into the surrounding water described in 3.7.10 within three seconds from water entry; and with temperatures pre-stabilized between -20°C (-4°F) and -65°C (-85°F) within three to seven seconds, the time increasing linearly with decreasing temperature. See Figure 7. (See i-1.6.H; I-2.Aa, Ab, Ac; I-3.Aa, Ab, Ac; IV-8)

* 3.8.7 Acoustic Output - The signal shall propagate a preselected coded sound, in the acoustic range, into the water, from "activation" to "termination". The sound shall be on one of two continuous wave frequencies, which shall be shift keyed, one to the other, to produce an intelligible code. The coded output within the tolerances specified: shall be continuous without break; shall shift frequencies on time; shall not vary or drift in frequency, amplitude, or pulse time due to increasing pressure, temperature variations, time, or power decay. The parameters and their tolerances for the signal output, are specified below. Acoustic terminology and measurements shall be in accordance with ANSI S1.20. (See I-2.Aa, Ab, Ac; I-3.Aa, Ab, Ac; IV-18)

(a) Operation - Output from "activation" to "termination" shall be at full power level for a period of 45 + 75 - 0 seconds. (See Figure 7) (See IV-9 & 10)

(b) Frequencies - The output shall be on the following frequencies (See Figure 8): (See IV-13 & 14)

(i) Fundamental: (See III-5)
f2 = 3.5 ± .05 kHz
f1 = 2.95 - 0 kHz and < f2 by 3% or more of f2
(f1 shall not be less than 2.95 kHz and shall be less than f2 by at least 3%)

(ii) 3rd Harmonic: (See III-6)
f2' = 10.5 ± .15 kHz
f1' = 8.85 - 0 kHz and < f2' by 3% or more of f2'
(f1' shall not be less than 8.85 kHz and shall be less than f2' by at least 3%)

* (c) Source Level - The source level of the fundamental frequency shall not be less than 160 dB//1μPa and the source level of the third harmonic frequencies shall not be less than 150 dB//1μPa at one meter throughout operation, as defined in 3.8.7(a), not including rise and decrease times during frequency shifting, 3.8.7(g). The source level shall be maintained throughout the pulse period, defined in 3.8.7(f). The fundamental source level shall be the arithmetic average of the f1 and f2 source levels, and the third harmonic source level shall be the arithmetic average of the f1' and f2' source levels. The source level of f1 and f2, and of f1' and f2' shall not differ by more than 3 dB in any pulse period throughout the operation, as defined in 3.8.7(a). During time key shifting, as defined in 3.8.7(h), the level of the switched off frequence shall be at least 40 dB below the source level of the transmitted frequency. The non-harmonic frequencies from the signal shall also be at least 40 dB below the source level of transmission. (III-2,3,5,6; IV-11,12,15)

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(d) Frequency Stability - Within any single pulse length, the frequencies f1 and f2 shall not vary more than 40Hz. The pulse length during which this requirement applies is defined as the completion of rise time to beginning of decay time for f2, and completion of decay time to beginning of rise time for f1; 3.8.5(g). (See IV-17)

(e) Directivity (See I-4)

(i) The transducer horizontal beam pattern shall be omnidirectional within ± 1 dB in any plane parallel with the transverse (horizontal) axis of the signal and perpendicular to the longitudinal (vertical) axis of the signal.

(ii) The transducer vertical beam pattern shall be omnidirectional within ± 1 dB within ± 60° of the transverse (horizontal) axis of the signal. At angles greater than ± 60° from the transverse axis, the vertical beam pattern shall be no greater than 4dB below the vertical beam pattern within ± 60° of the transverse axis.

(f) Pulse Period - The time from frequency shift keying to the subsequent shift shall be (See Figure 10): (See IV-16)

(i) Short: 0.5 ± 0.1 seconds (See III-4)

(ii) Long: 1.5 ± 0.2 seconds (See III-8)

(g) Frequency Shift - The rise time to f2 or f2' shall not exceed 0.01 seconds, and the time to decrease to f1 or f1' shall not exceed 0.05 seconds. See Figure 9. (See III-7, IV-19)

* (h) Code - The timed key shifting between frequencies shall be programmed as follows for each of the listed codes starting with f1 or f2, except Code 5 which shall remain f2 without shifting. See Figure 10. The f1 and f2 requirements also apply to f1' and f2', and "long" and "short" are pulse periods: (See II-5, IV-4)

Code 1 = f1 - long
f2 - long, repeated
Code 2 = f1 - short
f2 - long, repeated
Code 3 = f1 - short
f2 - short, repeated
Code 4 = f1 - long
f2 - short, repeated
Code 5 = f2 - continuous (MODs 1&2 only)

3.8.8 Termination - The signal, on completion of the coded acoustic output "time" shall terminate all output, and decay to zero within three seconds. The decaying output shall not change code and shall not start up again after once having shut down. See Figure 7. (See I-1.6.H; I-3.Aa, Ab, Ac; IV-9 & 10)

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4. QUALITY ASSURANCE PROVISIONS

* The contractor shall demonstrate that the signal and its packaging are ready for qualification. The Government then tests a sample, submitted by the contractor, in a laboratory, for performance in all the environmental extremes specified; and air drops a sample at sea, under SERVICE conditions. This qualifies the signal and its packaging, freezes the designs, and releases the contractor to production. A design record model is submitted by the contractor during qualification. The Government conducts a physical audit of the hardware against its data package, to establish the allocated baseline of the signal design at qualification. This is repeated, during production against production hardware, confirming that unauthorized changes have not been instituted. Preproduction testing and a data package are in accordance with the provisions of the contract. During production, the contractor conducts specified tests on all hardware, and more detailed tests on a sampling of all hardware; the Government conducts air drop lot sampling acceptance tests at sea under conditions then existing. Accepted lots are packaged. The packaging in turn is tested on a sampling basis before shipment.

4.1 Responsibility for Inspection - Unless otherwise specified in this specification, in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as required otherwise in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. No item shall be submitted until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.2 Classification of Inspections

(a) Preproduction

(i) Contractor Demonstration
(ii) First Article, (Government)
- Laboratory
- Air Drop
- Design Record Model (Retained by Government)
(iii) Physical Audit, Allocated Baseline and Configuration Control Data Evaluation (Government)

(b) Production

(i) Material receipt inspection (Contractor)
(ii) Individual tests (Contractor)
(iii) Sampling Tests (Contractor)
(iv) Lot Acceptance Tests (Government)
(v) Physical Audit, Production Baseline (Government)
(vi) Design Change (Government)

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* 4.3 Contractor Demonstration - The contractor shall demonstrate that the design is complete and in all respects ready for First Article testing by the Government. The tests in Table III shall be used for this purpose and shall include four air drops at sea from the high altitude portion of the envelope depicted by Figure 6 and two drops at the low altitude portion of this envelope. The low altitude portion may be at zero speed from a fixed structure. These airdrops are to demonstrate basic survivability and gross functioning; the selected code shall be intelligible and shall terminate as specified when monitored by an uncalibrated sonobuoy and receiver, or other sensor system (see 1.3). The Government First Article Test Activity, assigned by the contract, shall arrange for airdrop and monitoring services; and shall observe these tests and those conducted by the contractor.

* 4.4 First Article Testing - The contractor shall submit 67 signals and associated packaging in two increments for Government testing in accordance with the terms of the contract. The first increment shall be tested in Government laboratories and air drop facilities to determine that the design meets all the requirements of this specification to the extremes specified. Any test previously performed may be repeated. Table I shall be used as a guide for the laboratory phase. The First Article test activity may use all or part of Table I, in any sequence, or substitute any other equivalent tests to insure qualification of the design. The contractor shall provide a portion of the signals configured with monitoring and power leads and/or with the termination circuit or device made inoperative (3.8.8), as may be requested by the First Article test activity at the time of the contractor demonstration test (4.3). Spare power supplies shall likewise be supplied to enable repeating tests that encounter failures, not the fault of the units under test. Table IV shall be used as a guide for the air drop phase of the first increment. All defects must be corrected and part or all of the first increment repeated for confirmation prior to delivery of the second increment for the final air drop tests. The second increment shall successfully pass the requirements of Table IV without modification, change, or repair using an acceptance criteria of an AQL of 2.5% defective units for Major Defects and 6.5% defective units for combine Major and Minor defects. The second increment shall include five Design Record Models which will not be tested. The test shall be stopped, in either increment, at any point that accumulated defects are enough to warrant a report of failure, except when the test activity determines that it is in the best interest of the Government to continue. Acceptance, conditional acceptance, or rejection shall be in accordance with the terms of the contract. The Design Record Model delivery shall be contingent on successfully completing the air drop test of the second increment. The Design Record Model will be retained by the Government for a period equal to the storage shelf life, and will be used as the design reference, in addition to the configuration control data package, for the duration of the contract and as may be continued in follow-on contracts without repeating the First Article tests. The samples shall be in the quantities indicated in 6.4.

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4.5 Material Receipt - The parts and material to fabricate the signal shall conform to the parameters required in the contractor's data package representing the design approved at First Article. The contractor shall by inspection and testing qualify each alternate source prior to use. Changes in source of critical parts shall be qualified under change in design, see 4.11. Repeated failures, see 4.18.

4.6 Individual Tests - Each signal and packaging system component shall be inspected and/or tested as specified in Table II. All defects shall be corrected or the material rejected. Repeated failures, see 4.18.

* 4.7 Sampling Tests - Each lot of signals shall have a random sample selected, by the contractor, in accordance with MIL-STD-105, Level I. The contractor shall conduct the tests in Table III, and use an acceptance criteria of an AQL of 1.5% defective units for Major Defects and 2.5% defective units for combined Major and Minor Defects. Repeated failures, see 4.18. Signal sampling tests shall be conducted prior to delivery for Lot Acceptance Testing and packaging sampling tests shall be conducted prior to release for shipping.

* 4.8 Lot Acceptance Tests - The lots satisfactorily passing the signal sampling tests shall be submitted to the Government for air drop tests in accordance with Table IV. The sample size shall be in accordance with MIL-STD-105, Level I, single sampling plan for normal inspection (without tightened inspection). The acceptance criteria is an AQL of 4.0% defective units for Major defects and an AQL of 10% defective units for combined Major and Minor defects. Five spares shall be submitted with each sample to be used to replace units eliminated from the test. The remaining spares are required for surveillance testing in outgoing years.

* 4.9 Lot Rejection - Remaining spares, from rejected lots, will be returned to the contractor on request, at the contractor's expense; otherwise, they will be disposed of to scrap at termination of the contract. Rejected lots shall be reworked to eliminate all defects prior to resubmitting for testing. Rework plans and resubmission for acceptance testing require approval. Instructions for rework are subject to the provisions of the contract. See 6.2(i) and 6.7.1.

4.10 Physical Audit - The data package and/or its revisions, submitted under the provisions of the contract, may be evaluated by the Government to insure that the applicable environmental parameters of this specification are a requirement thereon, that all values have tolerances, and classification of defects are specified. The primary considerations shall be temperature extremes and high shock. The audit activity will compare the data package for the signal and its packing system against the hardware qualified under the First Article tests. Discrepancies shall be corrected prior to submittal of the

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first production lot. This shall be identified as the allocated baseline. Subsequent to the acceptance of the first production lot, another physical audit will be made against the production hardware, the data package, and the configuration control requirements of the contract. This shall be identified as the production baseline. Additionaly audits may be conducted consistent with lot rejection frequency, failure reports from the Fleet, and at the beginning of follow-on contracts without First Article testing.

* 4.11 Design Change - All design changes made during First Article will require retesting of all or portions of the First Article tests run prior to the introduction of the change. Submittal quantity of additional samples will be specified by the First Article test activity. The retesting will be on the same basis as resubmittal after a failure at First Article. Subsequent to First Article approval, all changes shall be in accordance with the Configuration Control provisions of the contract. All Class I changes, all design changes involving Critical Parts and Material, and all changes of source of Critical Parts and Material (not previously qualified) shall be qualified by part or all of the First Article test applicable to the change. The quantities and the extent of retesting shall be determined by the First Article test activity. All approved changes shall be incorporated in the Design Record Model, retained by the local Government representative, when they are implemented in production; and the changed part or component shall be provided to the other holders of the Design Record Model, 6.4(b)(ii).

* 4.12 Test Procedures - See 6.3

4.13 Records - The records of all tests conducted by the contractor shall be available on request by the Government. They shall be retained for the life of the contract and for any additional period specified by the Government representative or optionally may be turned over to the Government representative (DCAS) for storage until scheduled for destruction under the then existing records retirement requirements.

* 4.14 Standard Conditions - The following conditions are normal laboratory test conditions:

Air Temperature	+25° ± 5°C (+77° ± 9°F)
Water Temperature	+15° ± 5°C (+59° ± 9°F)
Humidity	20 to 88% R.H.
Sea Water	Available fresh water with Sodium Cloride added, 1.5 to 3.6% by weight.

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4.15 SUS MK 84 MOD 0 Test Provisions - The power supply compartment for the sea water battery (if so designed) is not closed off from the effect of the service environment and there is no provision for suspension; therefore, the following are exceptions to Tables I through IV:

Table I

1.6.Eb	Rain, deleted.
1.6.Fb	Vibration suspended, deleted.
1.6.Fc	Shock suspended, deleted.
1.6.Ld	Suspension Evaluation, deleted.
3.Aa thru Ac	Prepare sample by sealing and plugging for purpose of the test.
5	100 foot drop, delete.

Table II

3	Water entry plug and humidity indicator check, delete.
4	Suspension, delete

Table III

1	Seal test, delete.
2 thru 8	Prepare sample by sealing sea-water battery compartment off (if so designed) for the purposes of the test. A plastic bag or other device can be used; followed by a post test check for leakage.

Table IV

1.f	Bomb Rack, deleted.
Item 3	Indicator in power supply, delete.
Item 5	Suspension, deleted.

4.16 SUS MK 84 MOD 1 Test Provision - There is no provision for suspension; therefore, delete all reference to suspension from Tables I through IV; Table I, Sequence 1.6.Ld; Table II, Sequence 4; and Table IV Item 5.

4.17 SUS MK 84 MOD 2 Test Provision - Conduct all tests specified herein.

4.18 Repeated Failures - The contractor shall maintain a record of all parts and material defects that are identified in incoming inspection, and individual and sampling tests. Trends that indicate high failure rate or limited life shall require corrective action, such as change in source, tightened requirements, or a change in design. The trend record shall be available for inspection during the life of the contract.

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5. PACKAGING

* 5.1 General - The signal shall be preserved and packaged to Level A. Each individual signal power source shall be hermetically sealed, or the entire signal shall be hermetically sealed in a unit container. Six signals with hermetically sealed power sources or unit containers shall be packaged in an exterior container or pack. Twenty exterior containers shall be unitized on standard size pallets. The transportation and storage enviornments of the signal are also applicable to the preservation and packaging. The long term storage is under cover and the short term storage is unprotected in a ship-board or advanced base, and in aircraft environments. The Government may alternately provide (GFM) standard metal ammunition containers and pallets in accordance with the order, ready to use, or that re-quired refurbishment. The contractor is responsible for inspection that assures that each GFM item is ready to use (rust removed, painted, repairs made, and gaskets and missing parts replaced); and to provide all dunage, desiccant, humidity indicator, strapping and seals, and marking placards.

5.2 Parts and Material - The first objective in selection of parts and materials shall be the reliable fulfillment of the major performance objectives. The pallet shall conform to specified material requirements.

5.3 Design and Construction - The packaging shall be designed and constructed to perform as specified in the extremes of the trans-portation and storage environments. The shelf life shall equal that of the signal (3.8.1).

* 5.3.1 Unit Container - The unit container shall (not required with hermetically sealed signals):

(a) be hermetically sealed water and water/vapor tight (See I-1.4.G, III-10, IV-2)

* (b) have a full end panel tear out, for signal removal, that will not be hazardous to handle with normal caution accorded metal cans. (See I-1.4.I)

(c) have a continuous beaded or folded edge on open end (when panel is removed) to prevent cutting personnel. (See I-1.4.I)

* (d) incorporate anti-static treatment, if required, to prevent the buildup of an electrostatic charge in any of the materials used. (See I-1.4.K)

(e) have shock energy distributors in each end to protect signal and prevent panel failure.

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(f) have chafing protection as necessary to prevent removal of marking from signal due to vibration forces. (See I-1.4.K)

(g) be corrosion and fungus resistant on exterior and non-flammable. (See I-1.4.E & F, I-1.5.A)

* (h) be marked as specified in Figure 11. The marking color shall contrast the container exterior color, and shall have the same shelf life as the signal. Labels may be used, providing they are of a water and fungus resistant material that will survive the expected environments for the shelf life of the signal. Marking shall be 4.7mm (3/16 inch) high or greater; in Arabic numerals and Gothic or Futura or similar type letters. Labels or their adhesive shall not cause corrosion. (See I-1.4.B, III-11)

(i) be loaded with the signal in the same direction in all containers. (See I-1.4.K)

(j) be free from moisture inside the sealed container, that could, on condensation, cause failure of the signal's battery or cause corrosion or other deterioration of the signal or the inside of the container itself. Desiccant shielded from reaction with other material may be used, or other positive means such as a controlled atmosphere at sealing. (See I-1.4.K)

* (k) deleted

5.3.2 Exterior Container or Pack - The exterior container or pack shall:

* (a) hold six signals with hermetically sealed power sources or unit containers loaded with signals, the signals alternating to distribute equal loading if normally horizontal, or signals shall all be nose down if normally vertical. (See I-1.2.A, III-12 & 14)

* (b) either prevent entry of rain or be so constructed to completely drain off all rain, spray, or condensation that might accumulate. (See I-1.2.B, III-12)

* (c) provide shock, vibration, and damage protection to the signals with hermetically sealed power sources or unit containers containing signals. Damage shall be that which would be occasioned by tip-over onto other objects, and .92 meter (3 foot) handling drop. (See III-12 & 14, IV-1)

(d) support over-stacking of equal weight and volume containers up to 4.6 meters (15 feet) in height, including their supporting pallets. (See I-1.J)

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* (e) have a marking surface on one end and on adjacent side, and be marked as specified in Figure 11. The marking color shall contrast the container exterior color. The marking shall have a shelf life of 10 years; nine under cover and one year unprotected. The size shall not be less than 6.35mm (.25 inch) in height and shall be in Arabic numerals and Gothic or Futura or similar capital letters. Marking shall not cause corrosion of the container surface. Up arrows may be used where major shock protection has been concentrated in the bottom and/or stacking is restricted to one way. (See I-1.2.C & F, III-13)

(f) retain its specified stacking and protection capabilities while wet with rain or salt water. (See I-1.J)

* (g) be non-flammable when secured closed. Internal (non-exposed) protective dunage/liner shall be self-extinguishing and shall not produce toxic fumes in a fire that would not allow effective use of self-contained breathing apparatus in fighting that fire. Certification shall be provided to the Government at the time of First Article, specifying the fire-induced toxic and irritant properties of the material used in the packing and packaging; and the applicability of the Federal Hazardous Substance Labeling Act thereto, if applicable. (See I-1.1.A)

* (h) have interlocking top and bottom to prevent shifting on the pallet or in any subsequent storage on ship, aircraft, or ground handling equipment (only applicable if fabricated from metal or plastic). (See I-1.2.D)

(i) maintain single container stability at 30° incline as would occur on a ship, aircraft, or during handling on conveyors. (See I-1.2.D)

(j) include an instruction sheet or card. If the exterior container is non-watertight, either treat or protect the instructions against water damage, (6.3). (See I-1.2.J, III-14)

* (k) incorporate anti-static treatment, if required, to prevent the build-up of an electrostatic charge in any of the materials used. (See I-1.2.I)

5.3.3 Unitized Pallet Load - The pallet load shall:

* (a) be assembled in accordance with WR-54/131 (when GFM are Small Arms Ammunition Boxes MK1) or WR-54/167 (when GFM are 20mm Ammunition Boxes MK3 MOD3); or equivalent to WR-54/131 and/or WR-54/167 on a MK12 metal pallet fabricated in accordance with MIL-P-23312. The specified strapping, seals, and edge protectors shall be used. (See I-1.A, III-15)

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* (b) be marked in accordance with MIL-STD-129 using weather-resistant non-flammable fiberboard panels for marking in lieu of marking on the containers in the load or using tags; and in accordance with Figure 11. (See I-1.B, III-16)

5.4 Packaged Environment - The packaging, as described under 5.3 above, shall endure the environments specified for the signal, in its protected conditions. The packaging shall provide the protection. (See I-1.C)

5.5 Performance -

5.5.1 Unit Container Performance -

(a) The signal shall be protected from all moisture, sand, dust, dirt, and fungus. (See I-1.4.G, IV-2)

(b) The signal shall not come out of the container and the container shall not become unsealed during shock and vibration environments. (See I-1.4.A)

(c) Tip-over and 254mm (10 inch) free fall drops shall not unseal the container. (See I-1.4.C & D)

(d) Handling, without pressure, shall not collapse the container sides. (See I-1.4.H)

* (e) Signal removal shall be accomplished without the aid of tools that are not included. (Note: included tools are keys and pull lanyards or tape, or similar devices, assembled with the unit container and/or the signal). (See I-1.4.I)

5.5.2 Exterior Container or Pack Performance -

* (a) The exterior container or pack shall protect unit containers or signals with hermetically sealed power source: from crushing, from coming adrift from shock and vibration environments, from the results of tip-over onto other objects, from .92 meter (3 foot) free fall drops, and from the weight of stacking to a height of 4.6 meters (15 feet). Additionally, the signal with the hermetically sealed power source shall be protected from all moisture, sand, dust, dirt, and fungus. (See I-1.2.F & G, IV-1)

(b) The pack shall pass the "fire test, external source" specified in MIL-STD-648. (See I-1.1.A)

(c) One man shall be able to life and handle the assembled pack, at any point on the pack, without the pack coming apart, loosening, and/or spilling the contents. Handling shall be safe; and accomplished without cut hands or torn clothing. (See I-1.2.H, IV-1)

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(d) The exterior container shall be stable in free or unrestrained stacking a minimum of six containers high. (Note: If containers can only be stacked one way, there shall be up-arrows on all sides.) (See I-1.3.A)

(e) The exterior container or pack shall be capable of being opened with simple tools (or without) safely without cut hands, and tensioned components shall be closely restrained to prevent eye injury. (See I-1.2.I)

* (f) On opening the exterior container, the signals or unit containers shall be restrained from rolling free or falling. (See I-1.2.I)

5.5.3 Unitized Pallet Load Performance -

(a) The load (of exterior containers) shall be held together and not allowed to spill during and after: handling with fork lifts, being pushed, being towed, and being handled with pallet slings during underway and vertical replenishment. (See I-1.G & H)

(b) Level A rough handling shall not cause the load to loosen, making over-stacking unstable; nor shall it cause the load to come adrift. (See I-1.D, E, F, G)

(c) The top of the load shall not sustain damage that would affect performance (5.5.1, 5.5.2, and 5.5.3) from like loads superinposed on it during storage and shipment, including humping operations. (See I-1.K)

(d) Pallet loads shall be capable of remaining stacked up to 4.6 meters (15 feet) high, for up to 10 years, under cover.

(e) Pallet loads shall survive exposed deck cargo (one month), exposed land transporation (two weeks), and exposed field stowage (up to one year) environments and still be able to be handled without losing the load. (See I-1.C & I)

(f) Pallet loads shall not have parts or appendages arranged so as to be hazardous to handling personnel. (See I-1.L)

(g) Pallet loads shall be capable of disassembly with common tools and shall not be hazardous to personnel during breakdown. (See I-1.L)

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6. NOTES

* 6.1 Intended Use - See 1.1. This device is manually set to a code prior to loading in a dispenser, in a bomb rack adapter, in a sonobuoy launcher-container, on a bomb rack (when provided with suspension provision), or dropped into a free fall chute or from the door of a helicopter. The signal can also be manually prepared and launched from the surface. The signal propagates a time shift frequency coded sound in the water. The sound is received by a submarine. The coded sound is primarily an attack signal simulating a weapons drop and for one-way communications from an aircraft during submarine exercise. Other uses include: communications with swimmers, sound source for calibration or determining acoustic capabilities, and determining practice kill store misdistance by means of the Lloyd's Mirror Effect. The sound is detected by sonars and underwater telephone. Aircraft can confirm that the correct message was propagated by means of a radio link from sonobuoys in the water.

6.2 Procurement Requirements - The following is required:

		Ref
	(a) minimum qualification for contractor: (i) electronic underwater acoustic design and production (ii) aerodynamic and hydrodynamic design (iii) high impact shock design (iv) engineering test facilities, including: acoustic tanks; pressure, vibration, high shock, hot and cold temperature, time and frequency testing.	
	(b) Nomenclature, MARK and MOD	1.2
	(c) This specification number, revision and date, and each applicable amendment and its date.	
*	(d) The manufacturer's Data List or Parts List number, revision and date (applicable to follow-on contract) that identified the last approved production baseline for the signal and/or associated packaging as established by physical audits and change control procedures.	4.10
	(e) First Article Approval - See 6.4. (Not applicable in follow-on contracts)	4.4
*	(f) Quantity and lot size - Lot sizes shall be in full unitized (pallet) load increments in addition to test quantity and spares. Start-up lots shall be limited to two with the first having a quantity of 480 to be delivered plus 20 acceptance test units and 5 spares, and the second a quantity of 1080 plus 32 test units and 5 spares. Normal production lot sizes shall be 1920 plus 50 acceptance test units and 5 spares. The final lot quantity may be increased in increments of 6 (full box size) not to exceed 3198 without increasing the test quantity or spares.	4.8

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		Ref:
*	(g) Configuration Control requirement under MIL-STD-481.	4.11
*	(h) Designate/addresses of Government activities: (i) First Article Test Laboratory (ii) Air-drop Test Activity (iii) Configuration Audit Activity (iv) NAVAIRSYSCOMHQ Technical Code (v) Cognizant Field Activity	4.2
	(i) Assign "Rework Procedure" in 6.7.1 or an alternate for use.	4.9
*	(j) Assign "Precedence of documents" in 6.7.4 or an alternate for use.	
*	(k) Provide alternate packaging components as Government Furnished Material (GFM) and the requirement (if any) for refurbishment, as follows: (i) one MK12 pallet for each 120 signals to be delivered plus one for First Article tests. (ii) one Small Arms Ammunition Box MK1 MOD 0 or one 20mm Ammunition Box MK3 MOD 3 for each six signals to be delivered plus an additional quantity for all test and spare units as follows:	5.1 5.3.2 5.3.3 5.5.2 5.5.3

First Article, 1st Increment,	20	(on one pallet)
First Article, 2nd Increment,	6	
First Lot	85	(on four pallets)
Second Lot	187	(on nine pallets)
Subsequent (each)	330	(on sixteen pallets)

(l) Specify the delivery of the data required in 6.3.

6.3 Data Requirements - When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

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	Paragraph	Data Requirement	Applicable DID
*	4.10 3.6.12(b) 3.5.2(b)	Drawings, Engineering and Associated Lists Level 2. Include classification of defects in accordance with WR-43 or superseding publication. Single complete set. Preliminary non-reproducables 30 days prior to First Article to NAVAIRSYSCOM Technical Code, Cognizant Field Activity, First Article Test Laboratory, Configuration Audit Activity, and the local PCO representative. Microfilm in aperture cards with top legend completed, without tabulating cards or data deck, incorporating all changes that were required to pass First Article, 60 days prior to production, to the same addressees. New and revised microfilm at 120 day intervals thereafter. (New and revised only in follow-on contracts). Physical audit (4.10 of this specification) is required for approval.	DI-E-7031
*	4.12	Test procedures limited to Contractor conducted Individual and Sampling tests to NAVAIRSYSCOM Technical Code, Cognizant Field Activity, and the local PCO representative; Preliminary before Contractor Demonstration testing and Final 30 days prior to use in production; and when changes occur thereafter prior to use in production. The Cognizant Field Activity is the approving activity.	UDI-T-21347
*		Technical Manual. Change pages for NAVAIR 11-1-107 similar to Sec III of Chapter 4 therein with a requirement for MIL-M-38784 format, and review copies to NAVAIRSYSCOM Technical Code and the Cognizant Field Activity (CFA) prior to first production deliveries and approval by the CFA. (Not applicable in follow-on contracts).	UDI-M-21368
*	5.3.2(j)	Technical Manual. Instruction Sheet provided for Technical Manual NAVAIR 11-1-107, as illustrated therein, provides one per exterior container, with review copies to NAVAIRSYSCOM Technical Code and the Cognizant Field Activity (CFA) after First Article and approval by the CFA prior to use in production. Approval required in each contract.	UDI-M-21368

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Paragraph	Data Requirement	Applicable DID
4.6 4.7	Certificate of Compliance. Individual and Sampling testing, satisfactory completion of, each lot.	DI-E-2121
5.3.2	Certificate of Compliance. Packaging Material. Fire-induced toxic and irritant properties, and application of Federal Hazardous Substance Labeling Act. Provide to CFA and First Article test activity at First Article.	DI-E-2121
* 3.6.12(b) 4.4(a)	Data, Design - Equipment submitted for testing, limited to power supply replacement instructions to the First Article Test Activity and Cognizant Field Activity 30 days prior to First Article (Not applicable in follow-on contracts).	UDI-E-21338

(Copies of data item descriptions required by the contractors in accordance with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.4 First Article - When a First Article is required, it shall be tested and approved under the provisions of 7-104.55(b) of the Armed Services Procurement Regulation. The First Article should be a preproduction sample. The First Article should consist of the items listed below. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the First Article as indicated in 4.4.

(a) First Increment - Total 30 signals as follows:

(i) 6 signals, in unit containers (if power source not hermetically sealed), packaged in one exterior container, located in the bottom corner of a pallet load of exterior containers, each brought up to the weight and center of gravity of the first stated exterior container, for laboratory testing.

(ii) 16 signals, packaged for safe delivery for laboratory testing; some configured as specified at the time of Contractor Demonstration Testing, and including spare power supplies, desiccant, and humidity indicator (or spare signals with hermetically sealed power sources), also previously specified, for laboratory testing.

(iii) 8 signals, in unit containers (if power source not hermetically sealed, packaged for safe delivery, for preliminary air drop testing.

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* (b) Second Increment - Total 37 signals as follows:

(i) 32 signals, in unit containers (if power source not hermetically sealed), packaged in six exterior containers, for final air-drop testing.

(ii) 5 signals, marked "Design Record Model", packaged for safe delivery to be delivered subsequent to successful final air-drop testing, in the condition indicated, one each to:

- Government First Article Laboratory Test Activity (assembled)
- Physical Audit Activity (partially assembled) 1/
- NAVAIRSYSCOM Technical Code (assembled)
- Cognizant Field Activity (partially assembled) 1/
- Local Government Representative (partially assembled) 1/

* 6.5 Definitions - Hermetic: For purposes of this specification, hermetic shall be interpeted as water-vapor tight. This shall mean that no water vapor shall pass through or migrate through the body of the material, or through the seals in and around the material when subjected to vapor pressure 24.13mm (0.95 inches) of hg or 3208Nm^{-2} for 10 years (maximum and worst conditions).

6.6 Changes from previous issue - The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was down as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

6.7 Miscellaneous Notes -

6.7.1 Rework Procedure - (Applicable if specified in the contract) - Lots that fail acceptance tests shall be reworked and resubmitted for retesting. The following conditions shall be met before resubmission for acceptance:

(a) The cause for each defect that occurred during acceptance testing shall be determined, or the most probable causes shall be assigned each defect.

Note: 1/ Completed PC boards and preassembly wiring shall be provided and all parts, less cements and compounds, for the partially assembled configuration.

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(b) The Government representative shall make a random selection of 80 units from the remainder of the lot for screening. This screening sample shall be inspected/tested for each defect cause or probable defect cause that occurred during acceptance testing. REWORK/NO REWORK criteria for the entire lot based on the screening of thse 80 units is as follows:

(i) Any of these defects (paragraph (a)) found once (in the sample of 80), up to a total of three: the entire lot will not require rework before resubmission.

(ii) Any of these defects (paragraph (a)) found two or more times: the entire lot shall be reworked for each of these defects.

* (iii) Any of these defects (paragraph (a)) found once, adding to a total of four or more: the entire lot will be reworked for the defect or defects that have the greatest impact on reliable use (mission attainment) as determined by a review of defect history.

* (c) The rework procedure shall be submitted to the Government representative for approval. In the following cases, it shall be submitted to the Cognizant Field Activity for review and for procuring activity approval when: (1) the rework procedure justifies not reworking any specific defect requiring rework under the REWORK criteria in (b) above; (2) the procedure justifies recommendation to accept without airdrop testing in the cases that air dropping would not specifically test the defect or its cause; or (3) when no rework was to be performed under the NO REWORK criteria in (b) above.

(d) The entire lot shall be screened for the defect causes scheduled to be reworked. All signals found with these defect causes shall be reworked in accordance with the approved procedure.

(e) The lot shall be built up to the lot size specified in the contract and a random sample selected for acceptance testing.

* (f) Lots failing acceptance on the second submission may be resubmitted using the procedures in (a) through (e) above on approval of the procuring activity. Requests for approval of resubmission shall be addressed to the Cognizant Field Activity for recommendation and include the cause of each defect, the results of the re-screening, the rework procedure, and copies of the same for all prior submissions for that lot. Material, parts, or components reclaimed from lots failing resubmissions may be reused in accordance with procedures approved by the government representative.

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* 6.7.2 MODs - Figure 12 illustrates the basic MOD differences and the inter-relationship of the major components used in procurements prior to the revision of this specification. The basic configuration arrangement may have any functional/component/shape where not specifically specified such as: Switch in the tail, battery in nose, or a power source that has an isolated electrolyte that combines on shock of water entry, etc.

* 6.7.3 Water Entry Shock - The shock and concurrent contact surface pressures are extremely high when signals are released from high altitudes. Signal velocity is approximately 183m/sec (600 ft/sec) when released from 3048m (10,000 feet) and approaches 213m/sec (700 ft/sec) from 7620m (25,000 feet).

6.7.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specification are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications.

(c) Referenced Specification, Standards, and Requirements - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(d) Drawings - Drawings or lists of drawings referenced herein shall have precedence over all subsidiary drawings. All referenced drawings shall apply the extent specified thereon.

6.8 Cognizant Activity - NAVAIRSYSCOM (AIR-54901E).

Preparing Activity:
Navy - AS

Project 5845-NO63

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TABLE I. First Article laboratory tests.

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
* 1.A	Pallet Load complete, except one exterior container is loaded with 6 unit containers, each containing 1 signal, or 6 hermetically sealed signals without unit containers, all other exterior containers to weight and c.g. 1/	1	Examination that the assembly is in accordance with WR-54/131 or WR-54/167 or its equivalent	5.3.3(a)
1.B		1	Examination that marking complies with MIL-STD-129 and Figure 11	5.3.3(b)
1.C		1	Evaluation that materials will survive one year exposed stowage. Determination if materials tests will be required.	5.5.3(e) 5.4
1.D		1	Rotational drop, cornerwise: Fed-Test Method STD-101, Method 5005, Level A	5.5.3(b)
1.E		1	Rotational drop, edgewise: Fed-Test Method STD-101, Method 5008, Level A (optional)	5.5.3(b)
1.F		1	Free-Fall drop, flat: Fed-Test Method STD-101, Method 5007, Level A, Procedure B	5.5.3(b)
1.G		1	Impact, pendulum: Fed-Test Method STD-101, Method 5012, Level A, adjacent sides	5.5.3(a)&(b)
1.H		1	Mechanical handling: Fed-Test Method STD-101, Method 5011, forklift, slings, pushing, and towing	5.5.3(a)

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TABLE I. First Article laboratory tests. - Continued

	SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
	1.I	(continued)	1	Transportation: MIL-STD-810, Method 514.1, Procedure X, 3000 miles on land and 5,000 miles by air (optional)	3.7.1 3.7.4(a)(ii) 5.5.3(e)
	1.J		1	Superimposed Load: Fed-Test Method STD-101, Method 5016, weight to 15 foot stack of identical loads. Wet down at beginning of test	5.3.2(d) 5.3.2(f)
	1.K		1	Superimposed Load Shift: Weight to equal one like pallet load on a pallet; the superimposed load not secured by bracing, blocking, or tie-downs; duplicate freight rail car humping	5.5.3(c)
	1.L		1	Evaluate safety during load disassembly, using common tools	5.5.3(f)&(g)
*	1.1.A	Exterior Container with dummy load	1	Fire Test: MIL-STD-648. Evaluate Toxic and Irritant Certification and conduct material analysis, if warranted.	5.3.2(g) 5.5.2(b)
*	1.2.A	Exterior container load with either 6 hermetically sealed signals or 6 signals in hermetically sealed unit containers; from pallet load above 1/	1	Examination that load is distributed	5.3.2(a)
	1.2.B		1	Examination as to water shedding or drainage from superimposed load test	5.3.2(b)

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
1.2.C	(continued)	1	Examination that marking complies with surface, color, size, style, and Figure 11	5.3.2(e)
1.2.D		1	Evaluate interlock provision (if applicable)	5.3.2(h)
1.2.E		1	Test stability at 30° on conveyor in longitudinal and transverse attitudes	5.3.2(i)
* 1.2.F		1	Vibration (Repetitive Shock) Test: Fed Test Method STD-101, Method 5019, Bottom and one long side only if equipped with access cover, handles, and marked with up arrows; or all 6 sides if shape and design is not restrictive to normal handling and positioning.	3.7.2 5.3.2(e) 5.5.2(a)
1.2.G		1	Free-fall drop test: Fed-Test Method STD-101, Method 5007, Procedure C, 36" drop	5.5.2(a)
1.2.H		1	Handling evaluation (pick-up, carry, hand-to-hand, and stacking)	5.5.2(c)
* 1.2.I		1	Evaluate safety during container opening, and retention of unit containers; and using static charge measuring equipment that there is no electrostatic charge in any of the materials used.	5.3.2(k) 5.5.2(e)&(f)
1.2.J		1	Evaluate location, protection (if required), and contents of instruction sheet	5.3.2(j) 6.3

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
* 1.3.A	Exterior Containers with dummy load 1/	18	Free Stacking test, 6 high	5.5.2(d)
* 1.4.A	6 Unit Containers, each containing 1 signal (from the pallet load and exterior container 2/	6	Examine that signals are secure in containers and seal is not broken	5.5.1(b)
1.4.B		1	Examine that marking complies with color, size, style, and Figure 11	5.3.1(h)
1.4.C		2	Tipover test: Fed-Test Method STD-101, Method 5018	5.5.1(c)
1.4.D	(from tipover)	2	Drop Test (free-fall): Fed-Test Method STD-101, Method 5007, Procedure E modified to 10" drop, once on each end	5.5.1(c)
1.4.E		2	Corrosion Test: Fed-Test Method STD-101, Method 3007 (optional providing material is evaluated that it will survive one year exposure without loss of seal and legibility of marking)	5.3.1(g)
1.4.F		2	Fungus resistance test: Fed-Test Method STD-101, Method 3020 (optional providing material is evaluated that it will survive one year exposure without loss of legibility of marking)	5.3.1(g)
1.4.G		6	Sealing test: Fed-Test Method STD-101, Method 5009, Submersion Technique	5.3.1(a) 5.5.1(a)

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TABLE I. First Article laboratory tests. - Continued

	SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
	1.4.H	(continued)	1	Evaluate handling and fragility of container sides	5.5.1(d)
*	1.4.I		6	Evaluate opening containers safely without unattached tools, that containers edges are not sharper than normal food containers with pull-out end panels with a rolled end	5.3.1(b) 5.3.1(c) 5.5.1(e)
*	1.4.J	(deleted)			
*	1.4.K		6	Evaluate that all signals unload from their containers without the requirement for tools or devices not included, were all oriented in the same direction, that there is no moisture in the containers, that there is no indication that signal marking was damaged by chafing, and that there is no electrostatic charge in any of the materials	5.3.1(d) 5.3.1(f) 5.3.1(i) 5.3.1(j)
*	1.5.A	2 opened Unit containers (without signals) 2/	2	Fire Test: MIL-STD-648	5.3.1(g)
*	1.6.A	6 signals (from the pallet load and containers)	1	Form factor: Measure length, diameters, location of bearing surfaces; and evaluate code setting switch configuration. (See Figure 1)	3.6.1 3.6.9
*	1.6.B		6	Measure weight and location of c.g. for the record. (See Figure 1)	3.6.2 3.6.3
	1.6.C		1	Examination that marking complies with location, color, size, style and Figure 2	3.6.5
	1.6.D		6	Operate signals on all codes, on external power without causing termination. Use test receptacle. Reseal	3.6.6

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
* 1.6.Da	(continued)	2	Temperature shock: MIL-STD-810, Method 503. Check operation Temperature-Altitude: MIL-STD-810, Method 504 modified as follows, or other equivalent test method: STEP 1(at site, stabilize at +40°C (+104°F); STEP 2(increase altitude to 12,200 meters (40,000 ft) and hold 1 hour); STEP 3(reduce altitude to site and stabilize at -65°C (-85°F)); STEP 4(increase altitude to 12,200 meters (40,000 ft) and hold 8 hours); STEP 5(repeat steps 1-4); STEP 6(reduce altitude to site and stabilize at ambient, examine and check operation).	3.7.4(a)(i) 3.7.4(b)(ii) 3.7.4(b)(iv) 3.7.6(b)
1.6.Db		2	Fungus: MIL-STD-810, Method 508. 1 on side and 1 on nose. Check operation	3.7.4(b)(iii)
1.6.Dc		2	Humidity: MIL-STD-810, Method 507, Procedure I (maximum 50°C). 1 on side and 1 on nose. Check operation	3.7.4(b)(iii)
1.6.Ea	(from Fungus & Humidity)	2	Salt Fog: MIL-STD-810, Method 509, Procedure I, 1 on side and 1 on tail hook. Check switch and operation	3.7.4(b)(i)
1.6.Eb	(from Fungus & Humidity)	2	Rain: MIL-STD-810, Method 506, 1 on side and 1 on nose. Check operation	3.6.8 3.7.6(b) 3.7.8

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
	(continued)			
1.6.Ec	(from Temperature)	2	Dust: MIL-STD-810, Method 510, 1 on side and 1 on tail hook, all steps without break. Check switch and operation	3.7.4(b)(ii)
1.6.Fa		2	Free-fall Drop, 4ft, on hard surface, 1 on edge of nose and on side, and 1 on edge of tail. Check operation	3.7.5
1.6.Fb	(if applicable)	2	Vibration suspended horizontally on 14" spaced hooks, MIL-STD-810, Method 514.1, Procedure I, Curve M. Check operation	3.7.6 3.7.6(b)
1.6.Fc	(if applicable)	2	Shock suspended horizontally on 14" spaced hooks 14g's vertical, 11±1 millisecond, 3 times each. Check operation	3.7.6
* 1.6.G	(deleted)			
1.6.H		6	Operation to termination in manufactured sea-water with agitation, ambient temperatures, depth over 1ft. (Note: Manually remove battery flooding port closure, if applicable, prior to immersion). Note time to activation, termination, termination decay, and that there is no reactivation	3.8.6 3.8.8
1.6.I	(expended signals)	6	Examine marking for survivability and legibility	3.6.4
1.6.J	(expended signals)	6	Evaluate setting switch: Detent action, position, stops, index, code numbers, clockwise increase, likelihood of being damaged, means of setting, and setting with a red-lens flashlight for illumination	3.5(a) 3.6.9 3.6.5(k)

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
1.6.K	(continued) (expended signals)	6	Evaluate the human engineering, safety, and workmanship aspects	3.5(a) 3.6.13 3.6.15 3.6.16
1.6.La	(expended signals)	1	Materials evaluation; disassemble: electrolytic corrosion, leakage, fungus, embrittlement and other changes in properties that would affect life or usability, and that no explosive devices were used	3.5(b) 3.5(d) 3.5(j) 3.5(k)
1.6.Lb	(expended signals)	1	Fire Test: MIL-STD-648	3.5(g)
1.6.Lc	(expended signals)	2	Sink rate and underwater dispersion. (If performed in fresh water convert to sea water)	3.6.2 3.8.3
* 1.6.Ld	(expended signals) (if applicable)	2	Evaluate bomb rack fit, anti-sway torque resistance, clean drop away, and housing of suspension lugs. Bomb rack does not have to be in an aircraft. Evaluate any keeper pins or other devices used in loading and the provisions for their removal or disposal, including color and instructions. (Note: No parts, pieces, or devices are to be released separately from the signal).	3.6.5(p) 3.6.10 3.6.11 3.8.2

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
2.Aa	6 signals with 10ft monitoring wires to battery output and transducer input, termination circuit deactivated	2	Cold signals (one -20 ± 1°C and one -62 ± 2°C) operation in cold (1° ± 1°C) manufactured sea water, with agitation, depth over 1 foot. Test all acoustic parameters (except termination) for a period of 45 seconds: activation, frequencies, average output level, output on each frequency, pulse periods, shift times, and noise. 1 on code 1 and 1 on code 3.	3.7.6(b) 3.7.10 3.8.6 3.8.7
2.Ab		2	Warm signal (39° ± 1°C) in warm (35° ± 1°C) manufactured sea water, with agitation, depth over one foot. Test as above	3.7.6(b) 3.7.10 3.8.6 3.8.7
2.Ac		2	Ambient temperature signal in ambient temperature manufactured sea water, with agitation, depth over one foot. Test as above. Optional consistent with results of cold and warm tests	3.7.6(b) 3.7.10 3.8.6 3.8.7
* 3.Aa	6 signals	2	Cold signal, cold water (1° ± 1°C) manufactured sea water, signal stabilized in water at no more than 1 psig. Signal suspended on spring with solenoid to move signal in agitation. Power source water entry initiation device/ switch/seal secured or rigged to initiate on first agitation. At time water-entry start agitations and increase pressure in closed vessel at the rate of 6.4 ± 0.2psi per second continuing for 444 additional psig beyond termination time. Record activation time, average sound level and apparent changes in level, termination, and maintenance of no sound following termination.	3.8.3 3.8.6 3.8.7 3.8.8 3.5(c) 3.6.8

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TABLE I. First Article laboratory tests. - Continued

SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
3.Ab	(continued)	2	Warm Signal, warm water (35° ± 1°C) as above	3.8.3 3.8.6 3.8.7 3.8.8 3.5(c) 3.6.8
3.Ac		2	Ambient temperature signal and water as above. Optional consistent with cold and warm tests.	3.8.3 3.8.6 3.8.7 3.8.8 3.5(c) 3.6.8
3.B		6	Open to check water leakage. Optional except on failures	3.6.8
4	2 signals with 20 foot remote power supply leads and termination circuit deactivated. Tag with correct voltage, polarity, and impedance.	2	Determine directivity	3.8.7(e)
5	2 signals	2	Drop from a height of 100 feet into water. Release from a horizontal attitude. Drop from tower, crane, building, helicopter, or bridge. Minimum depth of water as recommended by contractor. The signal shall activate (in salt water) or its water entry port, for activation, shall have functioned; indicating that the signal would have functioned. Recovery lanyard (if used) shall be secured peripherally at c.g. and shall be paid out prior to drop to insure that it will cause minimum interference.	3.7.7

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TABLE I. First Article laboratory tests. - Continued

	SEQUENCE	SAMPLE CONFIGURATION	QTY	TEST/EXAMINATION (or equivalent to qualify in accordance with requirement)	REQUIREMENT
*	6	expended signals 3/	6	Replace power source and conduct functional test on each code with remote power supply at test point in accordance with refurbishing instructions provided by the Contractor	3.6.6 3.6.12 6.2(1)(vi)
*	7	refurbished signals with desiccant 3/	2	Determine Water Vapor Transmission Rate, compare against data package, and verify that the amount of desiccant specified is actually sufficient for one year exposure	3.6.8 3.6.12(b)
*	8	Humidity indicator 3/	1	Expose for 24 hours at 80 ± 2.5% R.H. at 23.9°C (75°F); observe activation. Confirm non-reversable in dry chamber with active desiccant, 10 days minimum.	3.6.12(b)

NOTES:

1/ GFM containers and pallets are not under test; Contractor furnished equivalent units are. The Contractor provided items are under test, such as: dunnage, strapping and seals, design, workmanship, inspection and test performance. Failure of GFM is not a Contractor defect, unless Contractor induced.

2/ Not applicable for units furnished without unit containers.

3/ Not applicable for units that are hermetically sealed. See 3.6.12(a) and 6.5

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TABLE II. Individual tests.

	SEQUENCE	CONFIGURATION	TEST/INSPECTION (all units produced receive all tests/inspections)	REQUIREMENT
*	1	Complete signal, less test point closure device, but with a temporary seal	Code select control will not come off with light finger pull	3.5(a)
	2		Check that all exterior assembly securing devices are in place and up tight or fully secured	3.6.16
*	3	(if applicable)	Twenty-four hours or more after complete assembly check that the humidity indicator has not been activated, and that gravity and handling inertia will not dislodge the power source seal by a guided 254mm (10") free-fall onto a hard Maple block in the direction of possible seal movement, the block relieved in the way of the seal	3.6.8 3.6.12(b) 3.7.5
	4	(if applicable)	Check that suspension lugs are fully housed under spring pressure by light test pull	3.6.10
*	5	Complete signal with external power hooked up. (Insure that external power is free from switch "on" and "off" transients that may defect the signal on conclusion of satisfactory test)	Operate on all codes. Insure that they are intelligible, that the code switch is detented in each position, and there is no output other than the code for that position or no output between detended positions.	3.6.9 3.8.7(h)
*	6	Complete signal hooked up to test equipment or otherwise made ready for compartment envelope integrity test(s).	Test both that section of the assembly that is to be moisture and precipitation tight prior to water entry, and that section of the assembly that is to be watertight during the underwater trajectory. Insure that the signal is resealed with an internal dry and inert atmosphere. (If applicable)	3.6.6 3.6.8

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TABLE II. Individual tests. - Continued

SEQUENCE	CONFIGURATION	TEST/INSPECTION (all units produced receive all tests/inspections)	REQUIREMENT
* 7	Complete signal with test receptacle sealed	Pass through right circular cylinder having an inside diameter of 3.062 + .010 - 0.0 inches and a minimum length of 15 inches; and lengthwise through two parallel 4 inch square plates spaced 15.002 + 0.010 - 0.0 inches apart. Roll signal 360° on flat surface or cylinder checking that no part of the Transducer Boot extends beyond nose and mid-body bearing surface line.	3.6.1(a) 3.6.1(b) 3.6.1(c) 3.6.1(d)

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TABLE III. Sampling tests.

	SEQUENCE	CONFIGURATION	TEST	MAJOR DEFECT	MINOR DEFECT	REQUIREMENT
*	1	Humidity indicator, random selection of a sample of one unit, from number drawn for use in lot prior to installation (N/A for hermetically sealed signals)	Expose for 24 hours at 80 ± 2.5% R.H. at 23.9°C (75°F); observe activation; and provide replacement. Verify not reversable in dry container with active desiccant, minimum 24 hours.	M110		3.6.12(b)
	2	Complete signal preconditioned by 4ft nose drop on soft wood surface (signal not to be damaged). Set on Code 3. Hook external power source to signal, seal as necessary. Place under water in acoustic measurement tank.	Measure average source level on fundamental. 160dB//1 Pa or greater	M100		3.8.7(c)
	3	(same)	Measure average source level on third harmonic. 150dB//1 Pa or greater	M101		3.8.7(c)
	4	(same)	Measure pulse period. 0.5 ± 0.1 seconds	M102		3.8.7(f)(i)
	5	(same)	Measure f1 and f2 frequencies, frequency difference, and level difference. (f1) 2.95kHz up to 3% of (f2) 3.5 ± .05 kHz	M103		3.8.7(b)(i) 3.8.7(c)

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TABLE III. Sampling tests. - Continued

SEQUENCE	CONFIGURATION	TEST	MAJOR DEFECT	MINOR DEFECT	REQUIREMENT
6	(same)	Measure f1' and f2' frequencies, frequency difference, and level difference. (f1') 8.85 up to 3% of (f2') 10.5 ± .15 kHz	M104		3.8.7(b)(ii) 3.8.7(c)
7	(same)	Measure frequency shift rise time and decay time. Rise 0.01 maximum and decay 0.05 maximum		M201	3.8.7(g)
8	(same) Set on Code 1	Measure pulse period. 1.5 ± 0.2 seconds	M105		3.8.7(f)(ii)
* 9	External power removed. Checked for leakage. Dry and resealed.	Check marking for correct data		M202	3.6.5
* 10	After lot acceptance and subsequent to signal sealing in unit containers select a reduced random sample. (Not used in contractor demonstration)	Conduct leak test: Fed-Test Method STD-101, Method 5009, hot water technique.	M106		5.3.1(a)
11	(same)	Check marking for correct data		M203	5.3.1(h)

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TABLE III. Sampling tests. - Continued

SEQUENCE	CONFIGURATION	TEST	MAJOR DEFECT	MINOR DEFECT	REQUIREMENT
12	Exterior shipping container complete with unit containers and signals. Use one from lot as sample. (Not used in contractor demonstration)	Check security	M107		5.3.2(a) 5.3.2(b) 5.3.2(c)
13	(same)	Check marking for correct data		M204	5.3.2(e)
* 14	(same) Open	Check number of unit containers, direction of signal for equal weight distribution, presence and location of dunnage or packing, and presence and protection of the instruction sheet, if applicable. Secure	M108		5.3.2(a) 5.3.2(c) 5.3.2(j)
15	Pallet load. Use one sample from lot. (Not used in contractor demonstration)	Check security	M109		5.3.3(a)
16		Check marking for correct data		M205	5.3.3(b)

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TABLE IV. Lot acceptance tests.

	SEQUENCE	TEST
	1.	PreFlight Inspection:
*	1.a	Conditions of exterior containers and marking
*	1.b	Conditions of unit containers and marking
*	1.c	Humidity indicator indication (if applicable)
	1.d	Serialize, set approximately equal number of signals on each code
*	1.e	Check each signal for marking and abnormalities, such as: cracks, cut transducer boots, unsafe conditions, test receptacle not sealed, switch not housed, water inlet port(s) (MODS 1&2) sealing device inoperative, etc.
*	1.f	Prepare for launching in all the options available in one flight: bomb rack, bomb rack adapter, SUS launcher, SLC with split liner and spacer, and free fall chute.
	2.	Flight Test:
*	2.a	Of the total sample, release the minimum number required to fail the lot at each of two altitudes 100 feet and 25,000 feet, the balance at any altitude/speed within the Figure 6 envelope. The 100 and 25,000 foot drops shall include a minimum of life duration, correct and intelligible code, and no reactivation observations. The balance shall include observations of all attributes under controlled conditions.
	2.b	Measure water entry, to activation, and to termination
	2.c	Measure average signal level on the fundamental and third harmonic and their f1-f2 (f1'-f2') signal level differences
	2.d	Measure frequencies on fundamental and their stability
	2.e	Measure pulse periods
	2.f	Evaluate code intelligibility

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TABLE IV. Lot acceptance tests. - Continued

	ITEM	PARAMETER	MAJOR DEFECT	MINOR DEFECT	REQUIREMENT
*	1	Exterior container shall protect unit containers and/or signal from normal shipping and handling damage		M200	5.3.2(c) 5.5.2(a) 5.5.2(c)
*	2	Unit container shall provide sealed protection for the signal (If applicable)		M201	5.3.1(a) 5.5.1(a)
*	3	The humidity indicator shall indicate a moisture free condition in non-hermetically sealed MOD 1&2 signals for a minimum of one year exposure		M202	3.6.12(b)
*	4	Code selected by switch shall be the code emitted by the signal without change to another code and the switch control shall have a positive detented feel at each code position	M100		3.6.9 3.8.7(h)
	5	Suspension lugs shall hold signal uptight to bomb rack and shall allow the signal to drop when the rack is activated (if applicable)	M101		3.6.10
*	6 & 7	Deleted			
	8	Activation time shall not exceed 3 seconds from water entry 1/	M102		3.8.6
	9	Termination time shall be in excess of 48 seconds from water entry	M103		3.8.7(a) 3.8.8
	10	Termination time shall not exceed 126 seconds from water entry		M205	3.8.7(a) 3.8.8

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TABLE IV. Lot acceptance tests. - Continued

ITEM	PARAMETER	MAJOR DEFECT	MINOR DEFECT	REQUIREMENT
11	Average output level for the fundamental frequency shall not be less than 160 dB// Pa	M104		3.8.7(c)
12	Average output level for the third harmonic shall not be less than 150 dB// Pa	M105		3.8.7(c)
13	Fundamental frequency for f1 shall not be less than 2.95 kHz and for f2 shall be 3.5 ± .05 kHz	M106		3.8.7(b)
14	The frequency difference between f1 and f2 shall exceed 3% of f2	M107		3.8.7(b)
15	The difference in f1 and f2 or f1' or f2' source level shall not exceed 3dB	M108		3.8.7(c)
16	The pulse period shall be 0.5 ± 0.1 and 1.5 ± 0.2 seconds	M109		3.8.7(f)
17	The frequency shall not vary more than 40Hz during f1 and f2		M206	3.8.7(d)
18	The code shall be intelligible and without breaks	M110		3.8.7
19	Frequency shift time from f1 to f2 shall not exceed 0.01 seconds, and from f2 to f1 shall not exceed 0.05 seconds.		M207	3.8.7(g)

NOTES:

1/ See Requirement for exception

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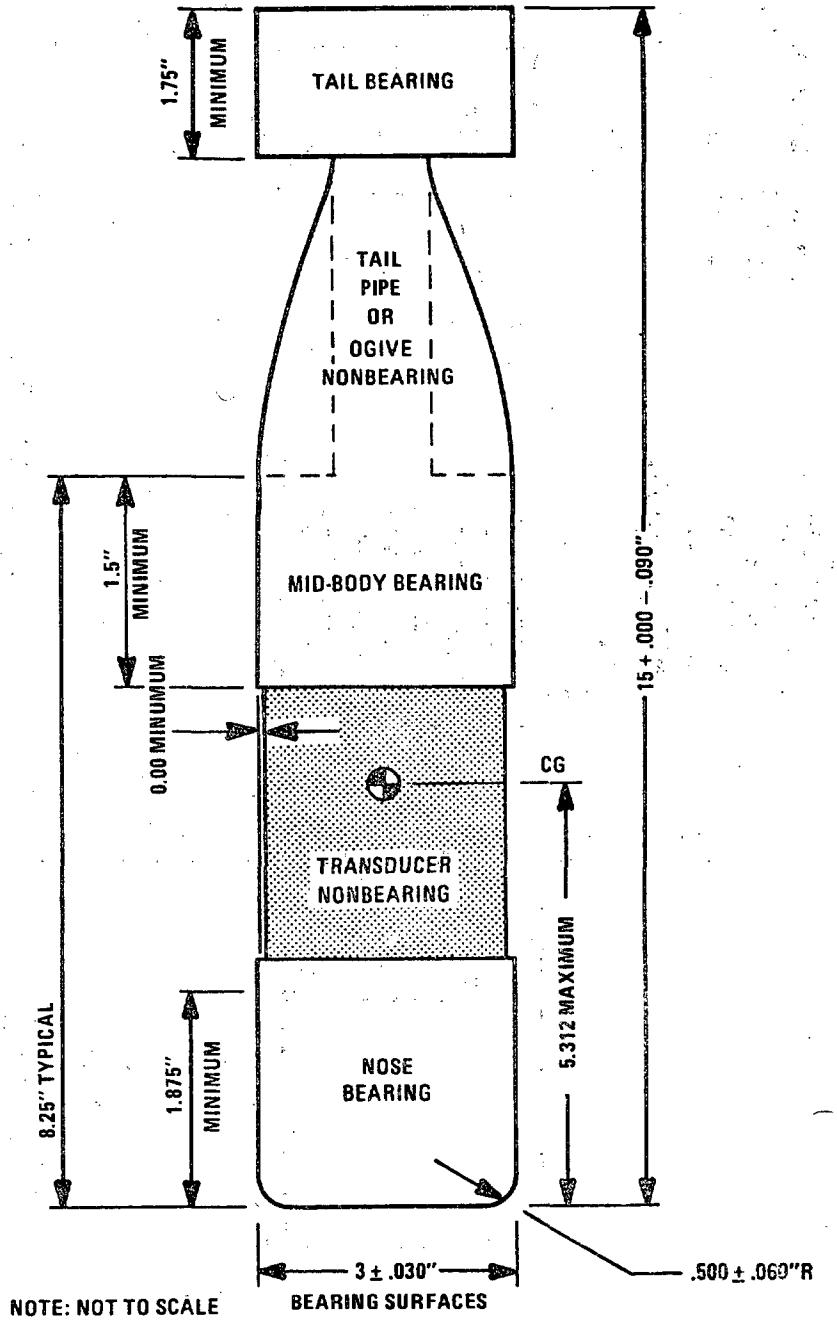
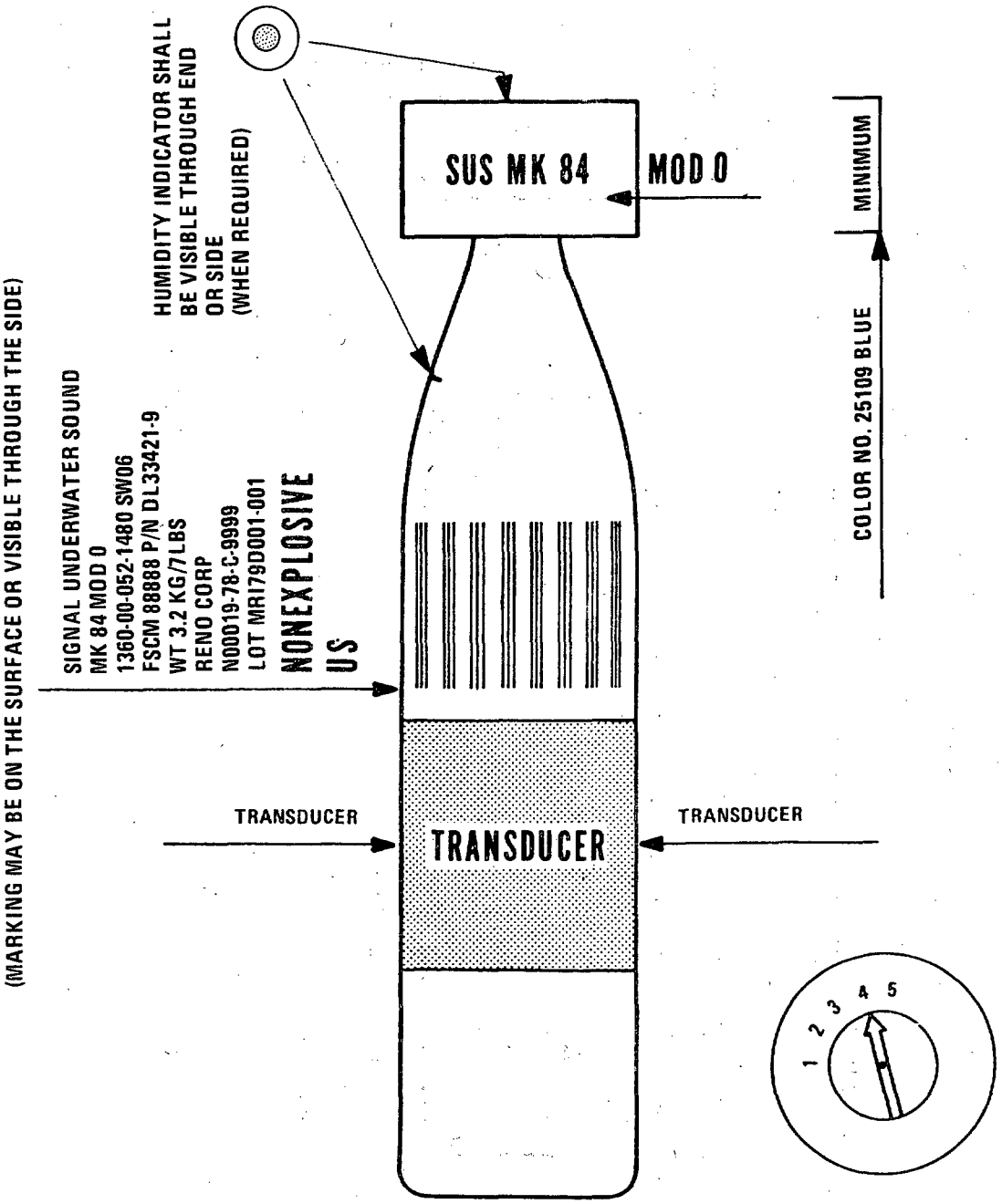


*Figure 1. Form factor and center of gravity.

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* Figure 2. SUS MK 84 marking.

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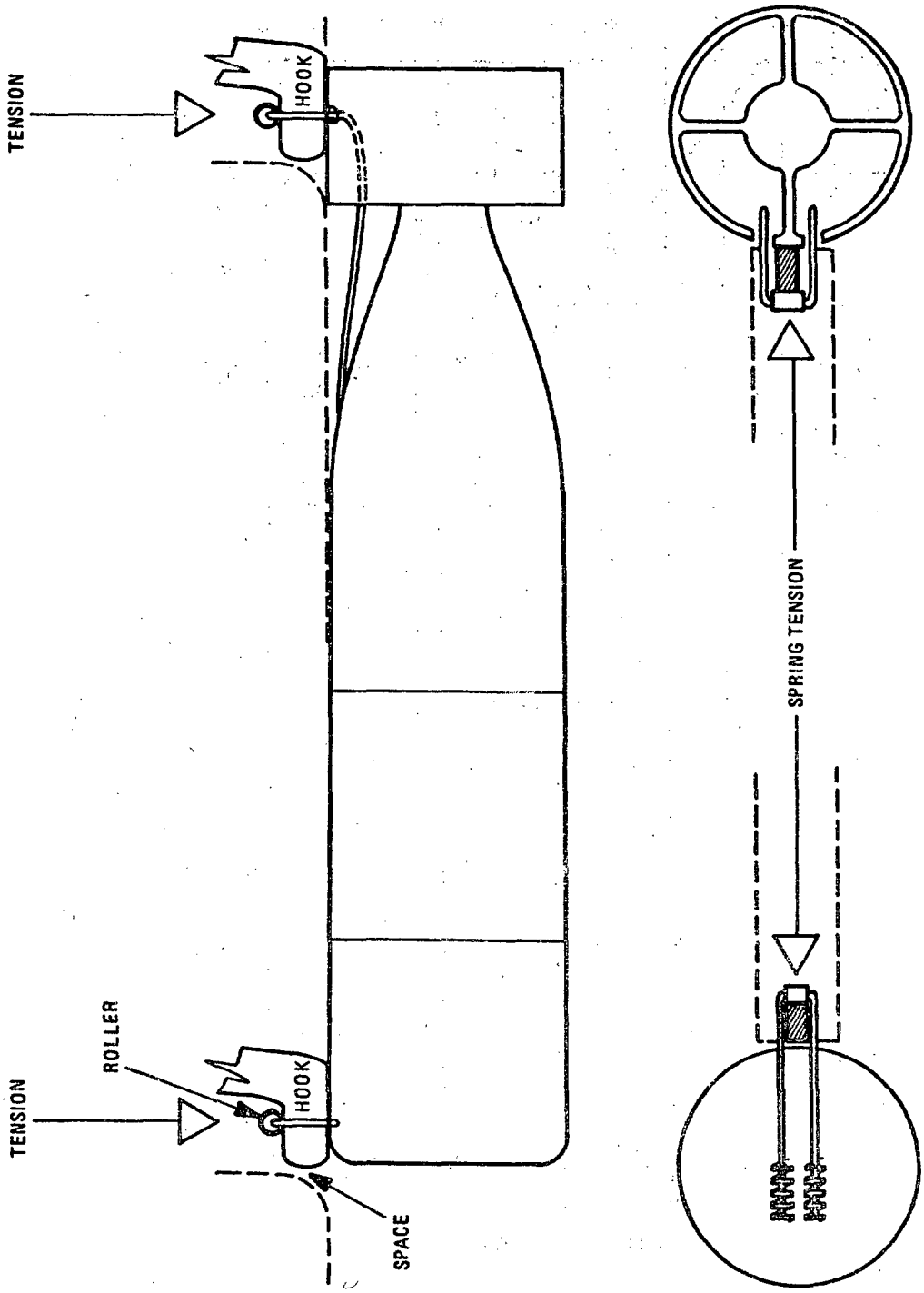


Figure 3. Suspension example.

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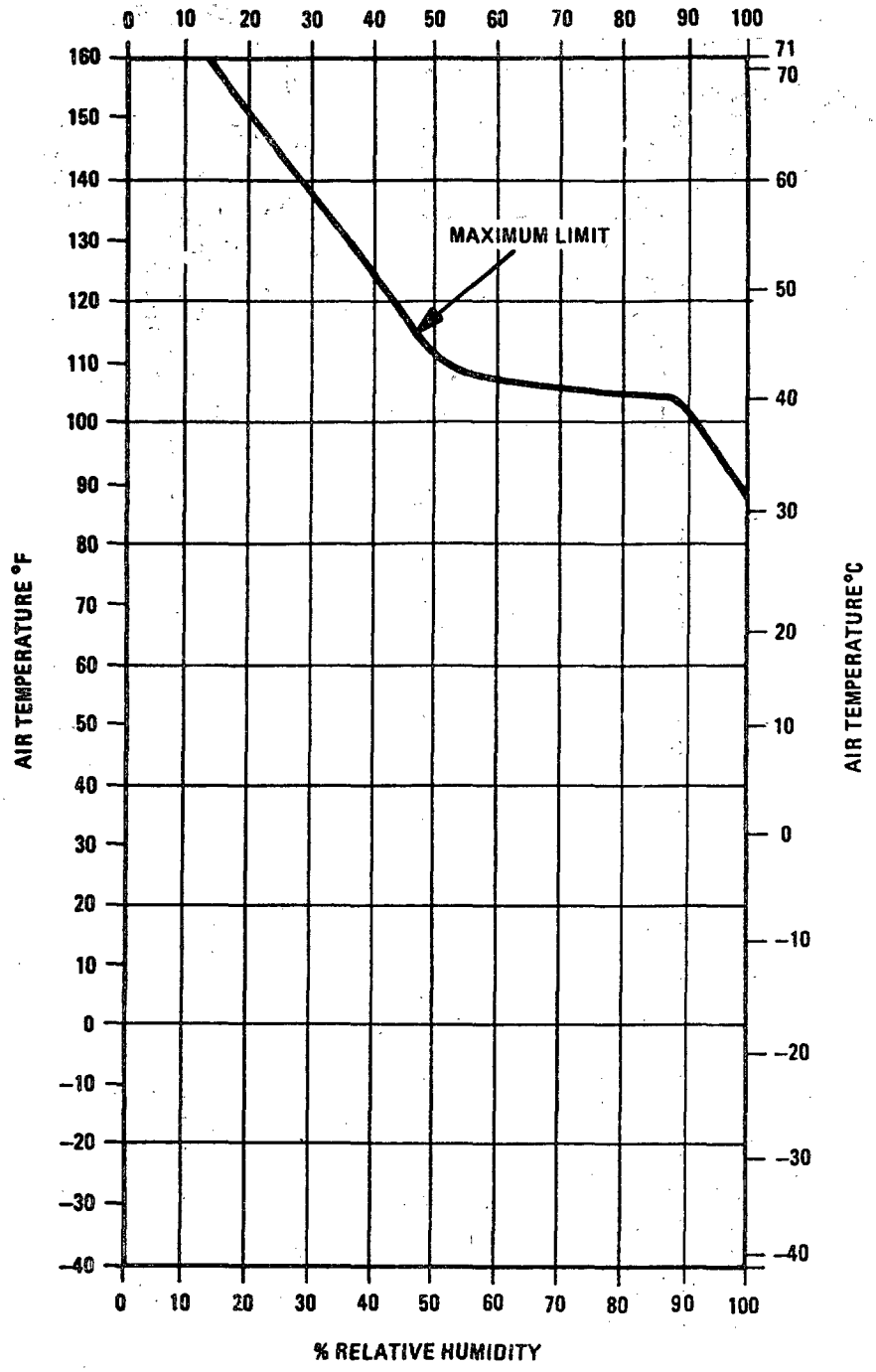
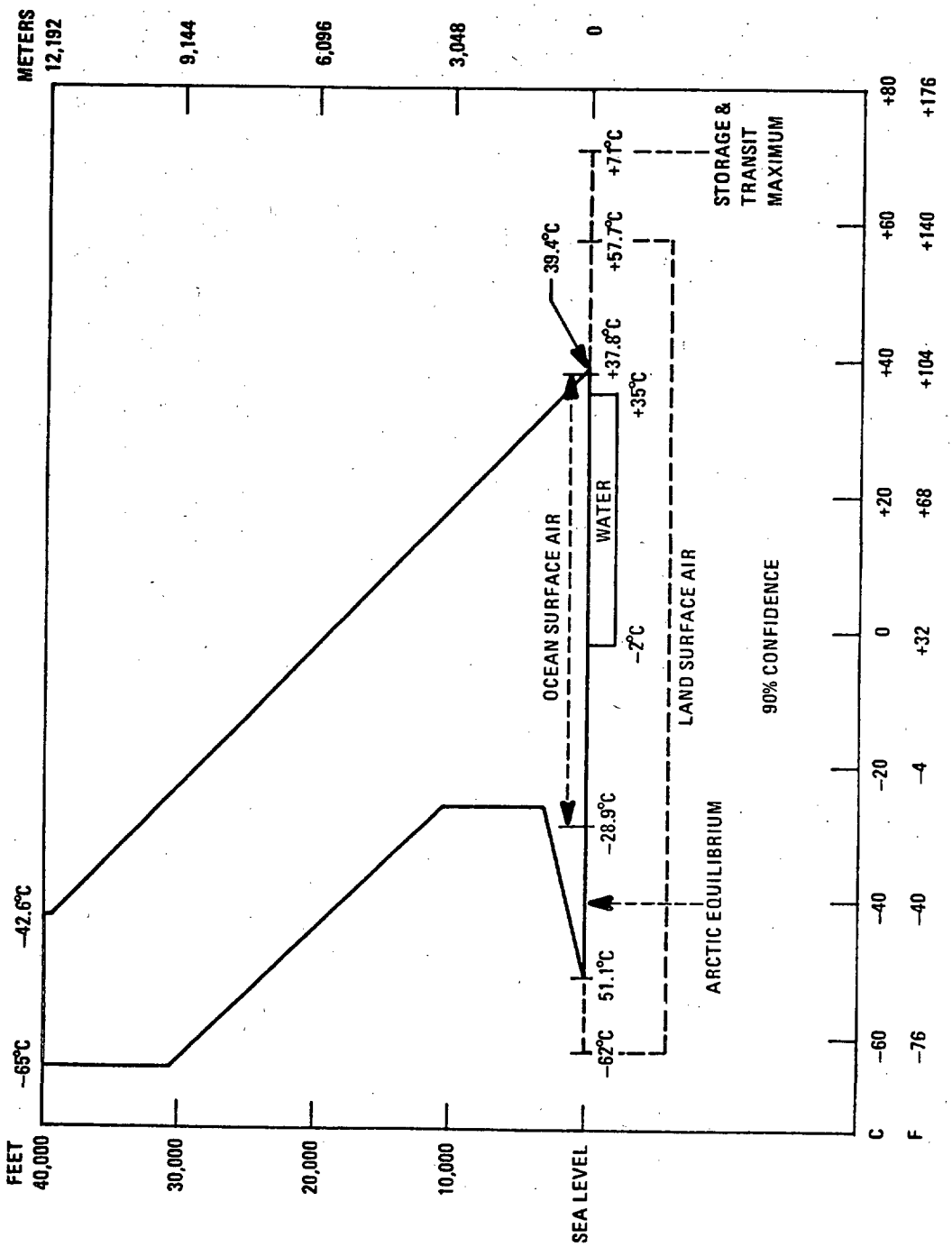


Figure 4. Relative temperature/humidity envelopes.

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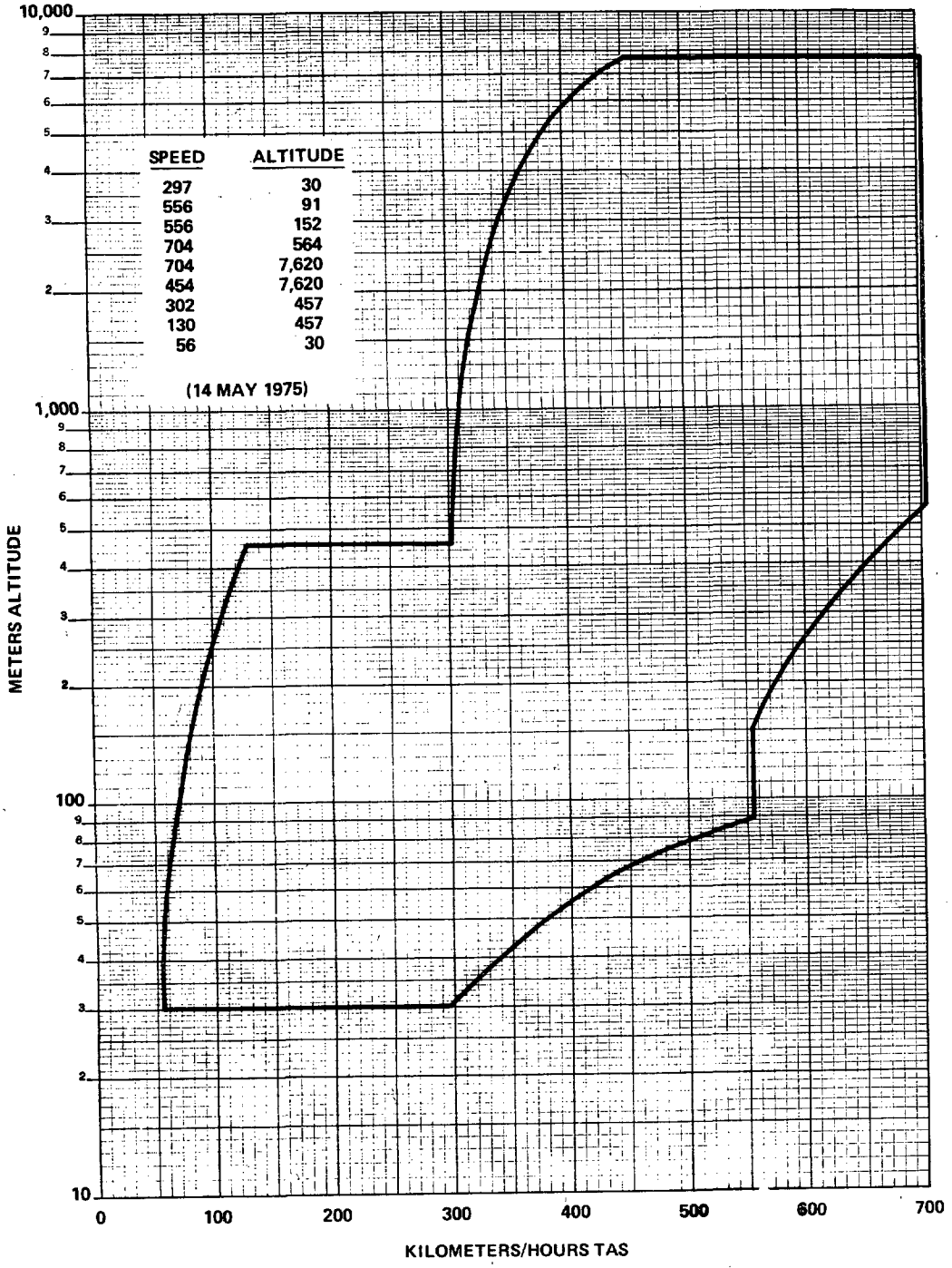


Figure 6a. Altitude/speed launch envelope.

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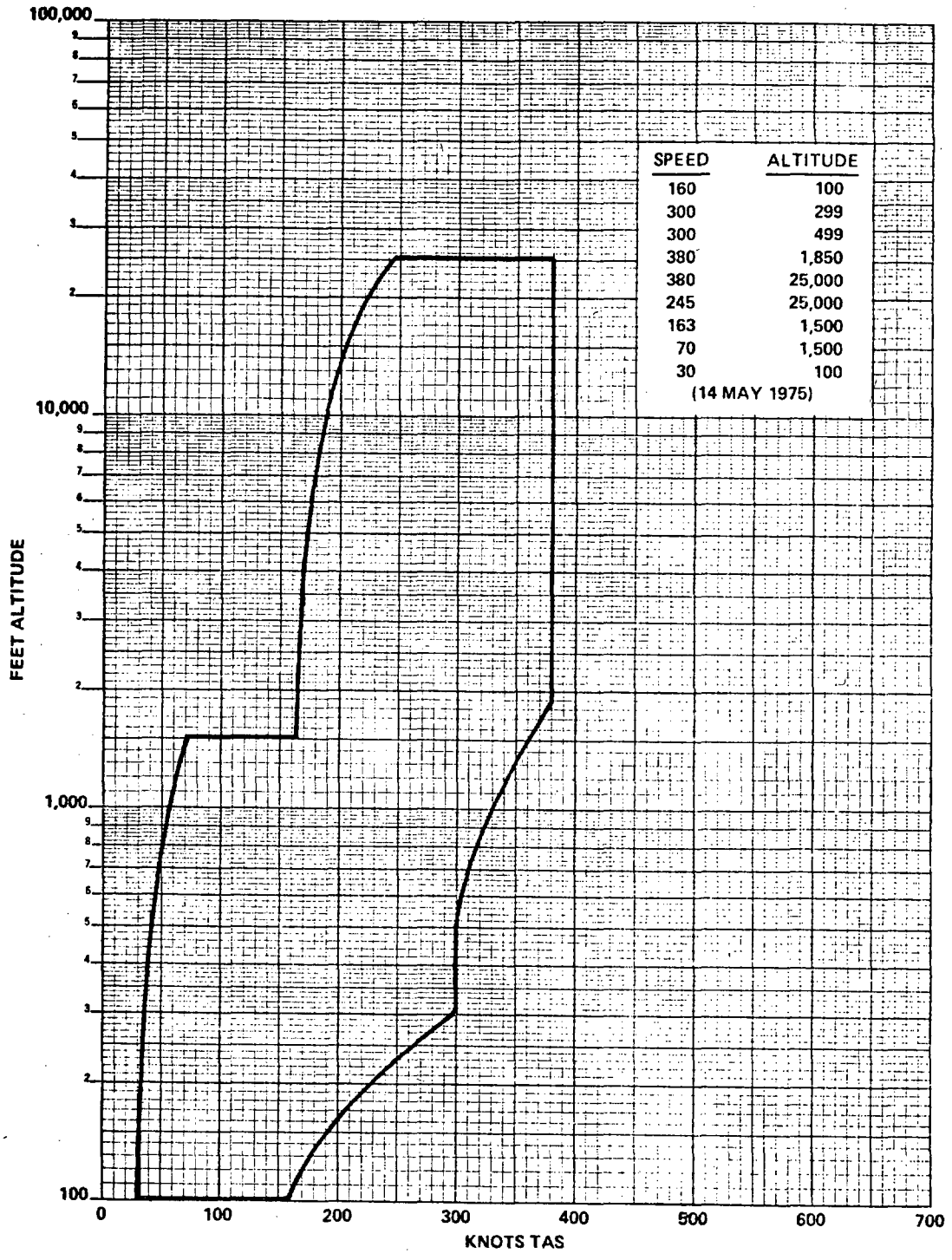


Figure 6b. Altitude/speed launch envelope.

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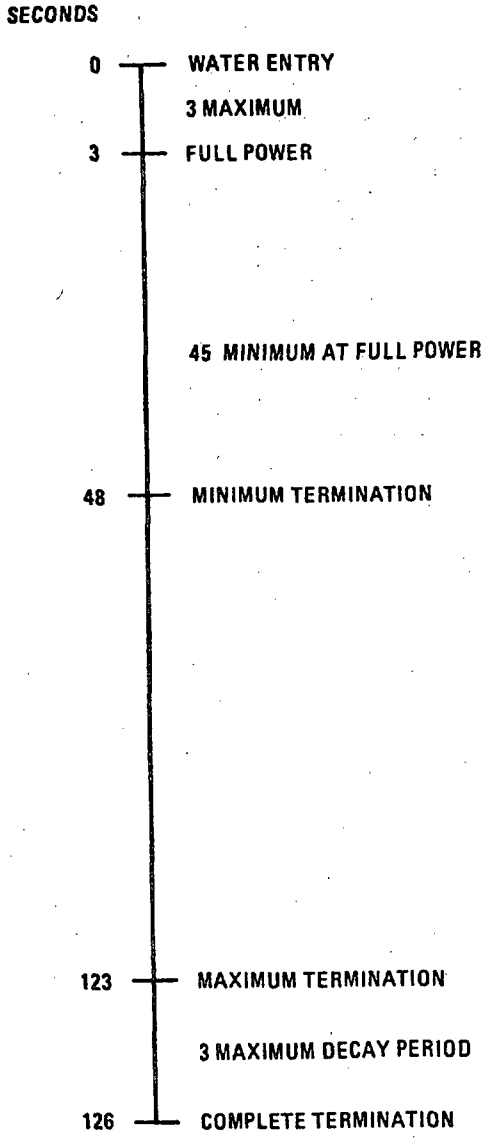
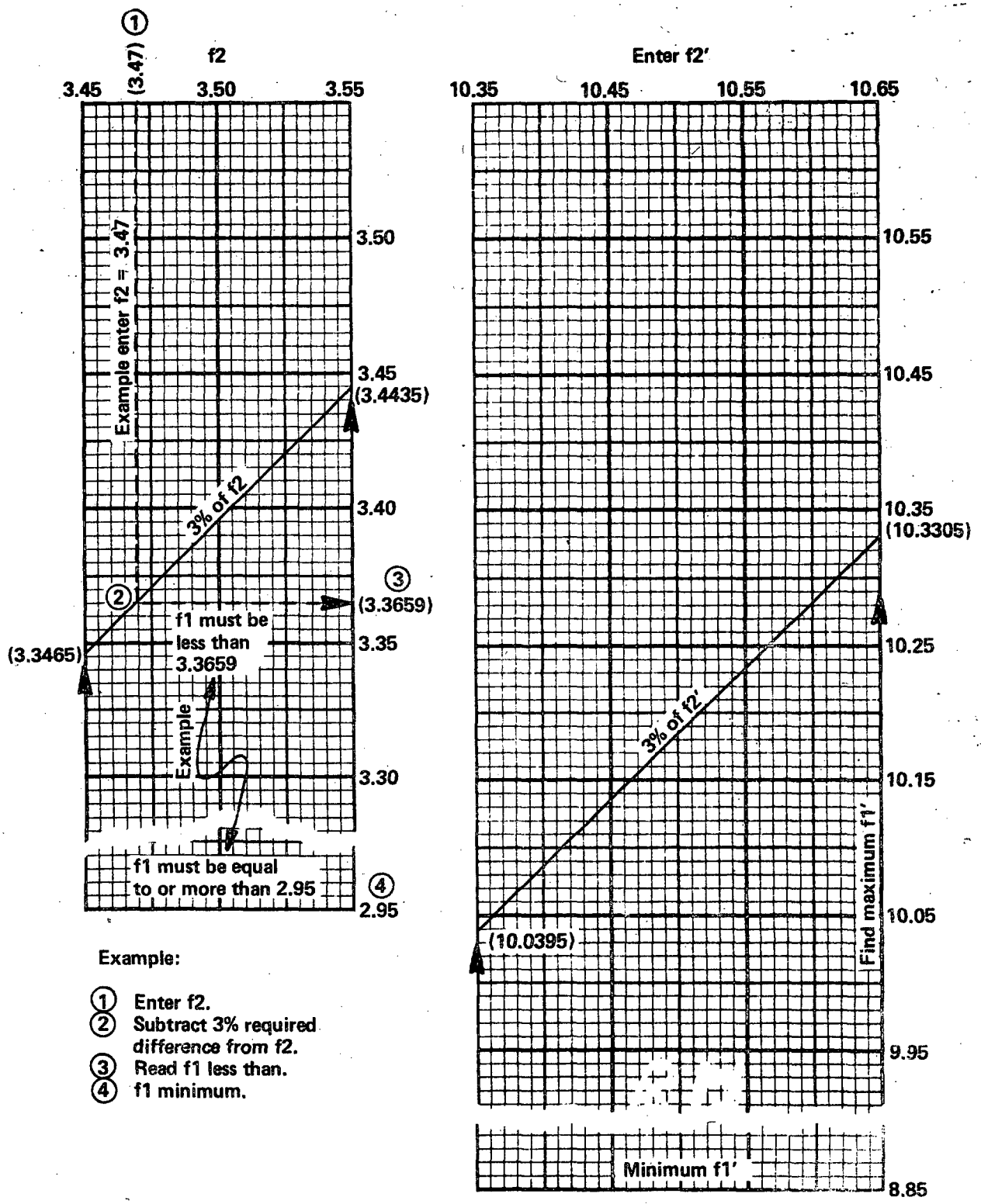


Figure 7. Operating time.

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* Figure 8. Frequency envelopes.

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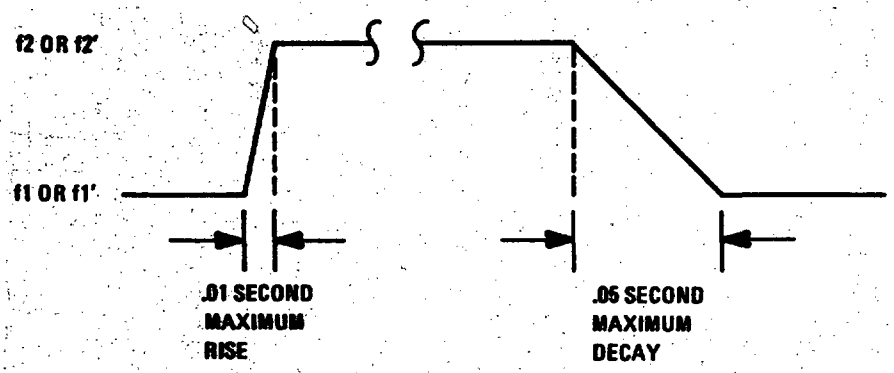


Figure 9. Pulse shift time.

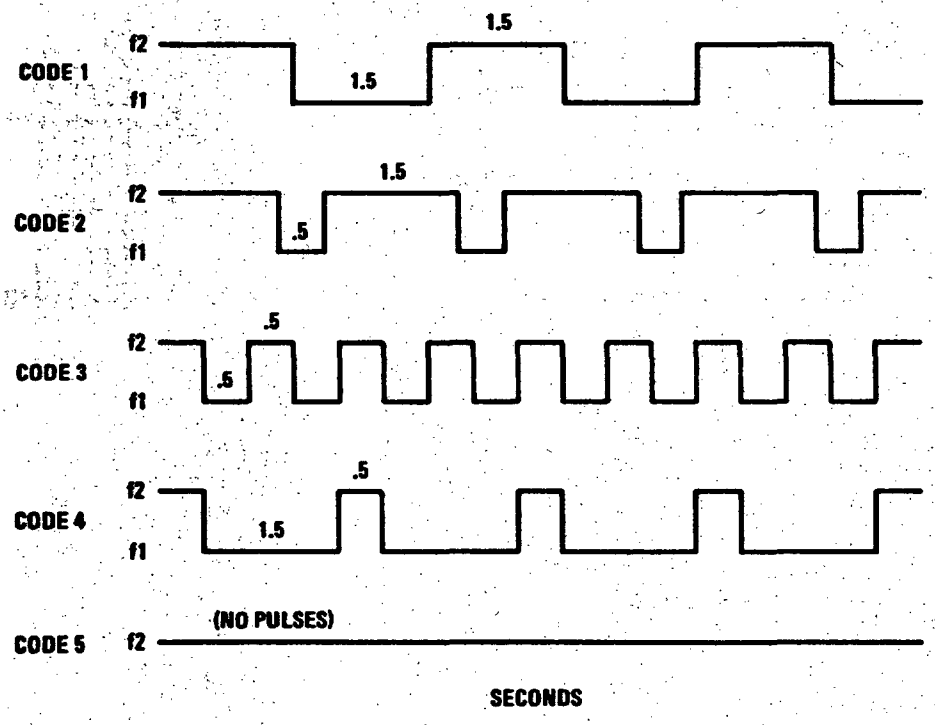
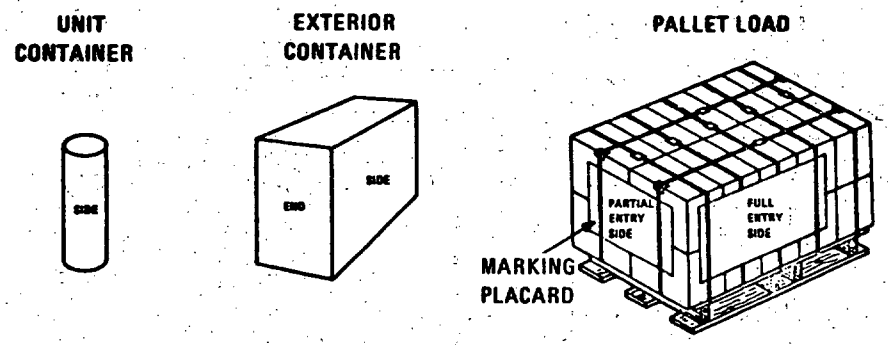


Figure 10. Codes.

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EXAMPLES:	UNIT CONTAINER	EXTERIOR CONTAINER		PALLET	
	SIDE	SIDE	END	FULL	PARTIAL
SIGNAL, UNDERWATER SOUND MK 84 MOD 0 (NONEXPLOSIVE)	X	X		X	X
SUS MK 84 MOD 0			X		
1360-00-052-1480 SW06	X	X	X	X	X
FSCM 88888 P/N DL334 21-9	X	X		X	X
(QUANTITY) EA	1	6	6	120	120
(LEVEL)	A	A/A		A/A	
WT	7 LBS	57 LBS		1250LBS	
CU	0.07	1.4		30.2	
LOT MRI79D001-001	X	X		X	
NAVAIR TEST DATE 0489 1/		X		X	
STACKING LIMIT 15 FEET				X	X
STORE UNDER COVER				X	X
HERMETICALLY SEALED DO NOT OPEN UNTIL NEEDED	X				
N00019-78-C-9999	X	X		X	
MANUFACTURER'S NAME AND ADDRESS	X	X		X	
PC NO 5/16				X	
(PACKING LIST ON PC)				1/16	

NOTE 1/ MIL-STD-129 TYPE II SHELF-LIFE

* Figure 11. Packing marking guide.

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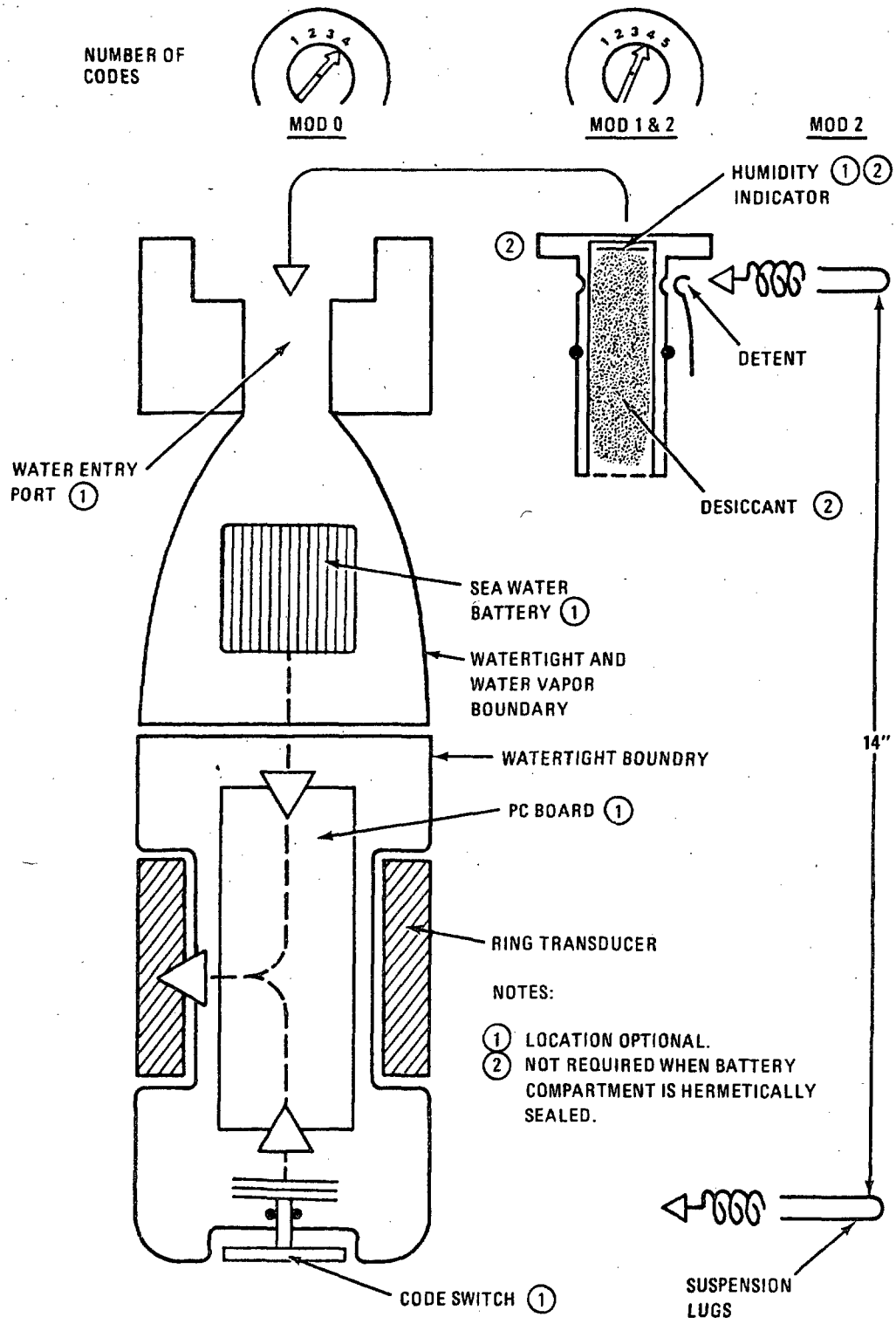


*Figure 12. Schematic, typical arrangement.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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